

The evolution of protostellar outflow opening angles and the implications for the Growth of Protostars

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ABSTRACT

We use 1 – 4 arcsec (300 – 1200 au) resolution ¹²CO (2 – 1) data from the MASSES (Mass Assembly of Stellar Systems and their Evolution with the Submillimeter Array) project to measure the projected opening angles of 46 protostellar outflows in the Perseus Molecular Cloud, 37 of which are measured with sufficiently high confidence to use in further analysis. We find that there is a statistically significant difference in the distributions of outflow opening angles for Classes 0 and I outflows, with a distinct lack of both wide-angle Class 0 outflows and highly collimated Class I outflows. Synthesizing our results with several previous studies, we find that outflows widen with age through the Class 0 stage but do not continue to widen in the Class I stage. The maximum projected opening angle reached is approximately $90^\circ \pm 20^\circ$, with the transition between widening and remaining constant occurring near the boundary between the Classes 0 and I phases of evolution. While the volume fractions occupied by these outflows are no more than a few tens of per cent of the total core volume, at most, recent theoretical work suggests outflows may still be capable of playing a central role in setting the low star formation efficiencies of 25 per cent–50 per cent observed on core scales.

Key words: methods: observational – techniques: interferometric – stars: formation – stars: low-mass – ISM: jets and outflows.

1 INTRODUCTION

Stars are assembled in molecular clouds through the gravitational collapse of dense cores of gas and dust (e.g. Shu, Adams & Lizano 1987; Di Francesco et al. 2007; Ward-Thompson et al. 2007; Dunham et al. 2014a; Pineda et al. 2023). The final masses of stars are set during the protostellar stage of star formation, when the forming protostar is embedded within, and accreting material from, its parent dense core. Andre, Ward-Thompson & Barsony (1993) divided this protostellar stage into Classes 0 and I protostars, where Class 0 protostars have accreted less than half of their final mass and Class I protostars have accreted more than half of their final mass. In practice, the observational class of a protostar is typically measured and assigned based on its observed spectral energy distribution (SED, e.g. Andre et al. 1993; Myers & Ladd 1993; Evans II et al. 2009; Frimann, Jørgensen & Haugbølle 2016), with the observational classes serving as proxies for the physical stages. In this paper, we follow the convention first introduced by Robitaille et al. (2006) of using ‘Class’

when referring to the observational categories and ‘Stage’ when referring to physical categories, although we note that Robitaille et al. used this terminology to distinguish between protostars and more evolved young stellar objects, and did not separately distinguish between Stages 0 and I protostars.

One of the specific methods of classifying protostars into their observational categories is based on the bolometric temperature, T_{bol} , defined to be the temperature of a blackbody with the same flux-weighted mean frequency as the SED of the object (Myers & Ladd 1993). It is in essence a protostellar equivalent of stellar effective temperature; it starts at very low values (~ 10 K) for cold, starless cores and eventually increases to the effective temperature of the central (proto)star once the core and disc have fully dissipated. Observational studies have placed the boundary between Classes 0 and I protostars at $T_{\text{bol}} = 70$ K (Chen et al. 1995).

A major unsolved problem in star formation is to understand the complex interactions of physical processes that set the final masses of stars, including mass infall, disc formation and growth, the formation of multiple systems through fragmentation on both core and disc scales, and mass loss through jets and outflows. While numerous studies have examined these processes, a complete

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understanding of the interplay between all of them remains lacking. The Mass Assembly of Stellar Systems and their Evolution with the Submillimeter Array (MASSSES) project represents an endeavour to study these physical processes using an unbiased sample of protostars with a homogeneous set of observations. MASSSES used the Submillimeter Array (SMA; Ho, Moran & Lo 2004) to map all of the protostars in the Perseus Molecular Cloud ($d = 300$ pc; Zucker et al. 2018, 2019) in the continuum and a variety of molecular line tracers at both 230 and 345 GHz. At 230 GHz, MASSSES used the combination of the subcompact and extended SMA configurations to achieve an angular resolution of approximately 3 arcsec (900 au at the distance to Perseus) while maintaining sensitivity to emission on scales up to approximately 20 arcsec (6000 au at the distance to Perseus). A full description of the MASSSES observations, data reduction, and public data releases are presented by Stephens et al. (2018, 2019). Other MASSSES papers to date have focused on the kinematics of protostellar cores (Lee et al. 2015; Heimsoth et al. 2022), the origins of fragmentation (Lee et al. 2015, 2016; Pokhrel et al. 2018), chemical evidence of an episodic accretion process (Frimann et al. 2017), the formation and growth of circumstellar discs (Andersen et al. 2019), and the connections between dense cores and the larger filamentary environments in which they form (Stephens et al. 2017).

In this paper, we use the MASSSES data to investigate the growth of protostellar outflows. Outflows from protostars may play a crucial role in regulating accretion of material onto stars by injecting momentum and energy into the surrounding medium and possibly feeding turbulence in the surrounding gas (e.g. Nakamura & Li 2007; Nakamura et al. 2011; Hansen et al. 2012; Plunkett et al. 2013; Frank et al. 2014; Offner & Arce 2014; Offner & Chaban 2017). Indeed, numerous studies have compared the dense core mass function (CMF) to the stellar initial mass function (IMF) and concluded that the IMF may be inherited from the CMF, with a star formation efficiency on dense core scales of approximately 25 per cent–50 per cent (e.g. Motte, Andre & Neri 1998; Motte & André 2001; Alves, Lombardi & Lada 2007; André et al. 2010; Könyves et al. 2015; Motte et al. 2022). In this scenario, outflows may be partly or mostly responsible for clearing the approximately 50 per cent–75 per cent of the dense core mass that does not become part of the final stellar mass, ultimately terminating the infall and setting the star formation efficiency on dense core scales (e.g. Velusamy & Langer 1998; Matzner & McKee 2000; Arce & Sargent 2006; Myers 2008; Machida & Hosokawa 2013).

If outflows are to remove a significant fraction of the initial core mass, they must be sufficiently wide to remove mass from, and prohibit accretion from, a substantial fraction of the solid angle of the core. In a seminal investigation of the opening angles of protostellar outflows, Arce & Sargent (2006) used ^{12}CO observations to show that outflows grow wider with time, reaching opening angles of approximately 180° by the end of the protostellar stage and thus halting further infall. Their results were reproduced in simulations (Offner et al. 2011), as well as in other studies measuring outflow cavity opening angles (Seale & Looney 2008; Velusamy, Langer & Thompson 2014; Hsieh, Lai & Belloche 2017; Hsieh et al. 2023). On the other hand, other studies have claimed that outflows are not capable of dispersing a large fraction of core material (e.g. Hatchell, Fuller & Richer 2007; Curtis et al. 2010; Habel et al. 2021). While the former two studies may be affected by underestimates of the total mass and energetics of the outflows in their sample (see e.g. Dunham et al. 2014b), the latter work by Habel et al. (2021) used infrared scattered light measurements of outflow cavity opening angles to argue against a widening trend with age.

In this work, we revisit the evolution of protostellar outflows using ^{12}CO data from the MASSSES project, leveraging the statistical power provided by the fact that MASSSES targeted the complete population of Classes 0 and I protostars in the Perseus Molecular Cloud. The organization of this paper is as follows: in Section 2, we briefly summarize the MASSSES sample and observational strategy, with references to previous MASSSES papers where these are described in more detail. In Section 3, we describe the fitting method we use to measure outflow opening angles. We present our results on the growth of protostellar outflow opening angles in Section 4. We then discuss our results in Section 5, including a detailed comparison to previous results (Section 5.1), a synthesis of what the combination of our results and previous results imply for the accretion of mass onto protostars (Section 5.2), and a discussion of the limitations of the current work (Section 5.3). Finally, we summarize our main findings in Section 6.

2 MASSSES SAMPLE AND OBSERVATIONAL STRATEGY

The MASSSES survey targeted all 74 known Classes 0 and I protostellar systems and candidate first hydrostatic cores within the Perseus Molecular Cloud as of 2014, when the survey was designed (Stephens et al. 2018, 2019). Perseus, located at a distance of ~ 300 pc (Zucker et al. 2018, 2019), was chosen for the MASSSES survey due to its relative proximity to Earth, its large number of Classes 0 and I protostars, and its declination of approximately $+30^\circ$ making it well suited for optimal SMA observations. MASSSES obtained continuum images at both 230 and 345 GHz, as well as image (spectral) cubes centred on a number of different molecular line transitions, in two different SMA array configurations, with spatial resolutions ranging from approximately 1–4 arcsec (300–1200 au at the distance to Perseus), depending on the image weighting used. Full details on the observational, calibration, and imaging strategies are presented by Stephens et al. (2018, 2019). All of the MASSSES data are now publicly available,¹ (see Stephens et al. 2019, for details). In this paper, we focus exclusively on the ^{12}CO (2 – 1) data, first presented in Stephens et al. (2018, 2019).

Galleries of all ^{12}CO (2 – 1) maps in the MASSSES data set are presented by Stephens et al. (2018, 2019). Since many of the protostars in Perseus are located in clustered regions, and individual maps centred on each pointing do not show the larger environment in which each protostar is located, we present in Appendix A large-scale maps showing the MASSSES ^{12}CO (2 – 1) images in the IC348, B1, NGC 1333, L1455, and L1448 regions of Perseus. These are not mosaics, rather they plot the individual pointings on one large map, offset from each other to reflect their positions relative to each other.

In total, we are able to measure outflow opening angles for 46 different outflows in the MASSSES sample, using a method we describe below in Section 3. Table 1 reports the name of each driving source (along with common names frequently used for each source), the clustered region of Perseus in which this protostar is located (if any), the position of the protostar as traced by our MASSSES 1.3 mm continuum detections when available (Pokhrel et al. 2018; Stephens et al. 2018), or by the VLA Nascent Disk And Multiplicity Survey of Perseus Protostars (VANDAM; Tobin et al. 2016, 2018), the bolometric temperatures of the protostars as compiled by the eHOPS (Extension of the Herschel Orion Protostar Survey out to

¹See https://dataverse.harvard.edu/dataverse/full_MASSSES/

Table 1. MASSES protostars with measured outflow opening angles.

Source	Common name	Region ^a	Protostar RA ^b (J2000)	Protostar Dec. ^b (J2000)	T_{bol}^c (K)	Robust	Beam major and minor axes (arcsec) ^d	v_{systemic} (km s ⁻¹) ^e	v_{blue} (km s ⁻¹)	v_{red} (km s ⁻¹)
Per-emb 1	HH211-MMS	IC348	03 43 56.770	+32 00 49.865	24.1	-1	1.6×1.2	9.4	2.0-5.0	13.0-16.0
Per-emb 2	IRAS 03292+3039		03 32 17.915	+30 49 48.033	30.9	+1	2.8×2.6	7.0	2.0-5.0	9.0-12.0
Per-emb 3		NGC 1333	03 29 00.554	+31 11 59.849	33.8	+1	3.4×2.0	7.3	-2.0 to 2.0	10.0-14.0
Per-emb 5	IRAS 03282+3035		03 31 20.931	+30 45 30.334	35.9	+1	2.8×2.6	7.3	-1.0 to 4.0	10.0-15.0
Per-emb 6		B1	03 33 14.404	+31 07 10.715	75.8	+1	1.9×1.8	6.2	2.0-4.5	7.5-10.0
Per-emb 7			03 30 32.681	+30 26 26.480	35.3	+1	2.9×2.2	6.5	2.0-5.0	7.0-10.0
Per-emb 9	IRAS 03267+3128	NGC 1333	03 29 51.876	+31 39 05.516	37.9	-1	1.7×1.3	8.2	1.5-5.5	11.5-15.5
Per-emb 10		B1	03 33 16.412	+31 06 52.384	34.6	-1	1.3×1.0	6.4	2.0-5.5	8.5-12.0
Per-emb 11	IC348MMS	IC348	03 43 57.055	+32 03 04.669	34.4	+1	2.0×1.9	9.0	2.5-5.0	12.0-14.5
Per-emb 13	NGC 1333 IRAS4B	NGC 1333	03 29 11.990	+31 13 08.140	25.0	-1	1.5×1.3	7.1	-1.0 to 2.0	12.0-15.0
Per-emb 15	RNO15-FIR	NGC 1333	03 29 04.190	+31 14 48.430	51.5	+1	4.7×3.3	6.8	4.0-5.0	11.0-12.0
Per-emb 16		IC348	03 43 50.999	+32 03 23.858	37.4	-1	1.1×0.9	8.8	5.5-7.0	10.0-11.5
Per-emb 17	IRAS 03245 + 3002	L1455	03 27 39.120	+30 13 02.526	54.8	+1	2.2×1.8	6.0	0.0-3.0	9.0-12.0
Per-emb 19		NGC 1333	03 29 23.498	+31 33 29.173	59.5	+1	2.9×2.7	7.8	4.5-7.0	10.0-12.5
Per-emb 20	L1455-IRS4	L1455	03 27 43.199	+30 12 28.962	62.1	+1	2.4×1.9	5.3	0.0-3.0	7.0-10.0
Per-emb 21		NGC 1333	03 29 10.690	+31 18 20.150	56.5	+1	1.9×1.5	9.0	-2.0 to 5.0	11.0-18.0
Per-emb 22	L1448-IRS2		03 25 22.353	+30 45 13.213	43.9	+1	3.2×2.7	4.3	3.5-6.5	6.0-12.0
Per-emb 23	ASR 30	NGC 1333	03 29 17.250	+31 27 46.340	41.7	-1	1.4×1.1	7.8	3.5-6.5	10.5-13.5
Per-emb 24		NGC 1333	03 28 45.297	+31 05 41.693	67.0	+1	2.5×2.0	7.8	0.5-3.0	12.0-14.5
Per-emb 25	IRAS 03235+3004		03 26 37.490	+30 15 27.900	63.4	+1	2.0×1.3	5.8	3.0-4.0	6.0-7.0
Per-emb 26	L1448C, L1448-mm		03 25 38.870	+30 44 05.300	41.9	+1	1.9×1.8	5.4	-5.0 to 0.0	10.0-15.0
Per-emb 27	NGC 1333 IRAS2A	NGC 1333	03 28 55.560	+31 14 37.170	50.2	+1	2.1×2.1	8.1	1.5-4.5	10.5-13.5
Per-emb 28		IC348	03 43 50.987	+32 03 07.967	59.2	+1	2.1×2.0	8.6	5.5-7.5	9.5-11.5
Per-emb 29	B1-c	B1	03 33 17.860	+31 09 32.307	47.4	+1	3.4×2.8	6.4	0.5-4.5	8.5-12.5
Per-emb 30	HH789	B1	03 33 27.330	+31 07 10.290	96.9	+1	3.7×3.1	6.9	1.0-4.0	10.0-13.0
Per-emb 31	[JJK2007]9	NGC 1333	03 28 32.547	+31 11 05.151	76.3	+1	2.7×2.3	8.0	2.5-6.5	10.5-14.5
Per-emb 34	IRAS 03271+3013		03 30 15.190	+30 23 49.110	97.6	+1	3.6×2.9	5.7	1.0-4.0	8.0-11.0
Per-emb 36	NGC 1333 IRAS2B	NGC 1333	03 28 57.360	+31 14 15.610	95.1	+1	3.4×3.0	6.9	-3.0 to 3.0	11.0-17.0
Per-emb 40	B1-a, IRAS 03301+3057	B1	03 33 16.650	+31 07 54.810	128.9	+1	3.3×2.8	7.4	1.5-4.0	9.0-11.5
Per-emb 41	B1-b	B1	03 33 20.341	+31 07 21.355	212.6	+1	3.4×2.8	6.5	1.0-4.0	8.0-11.0
Per-emb 42	L1448C-S		03 25 39.135	+30 43 57.909	48.8	+1	1.9×1.8	5.8	2.0-3.0	7.0-8.0
Per-emb 44	SVS 13A	NGC 1333	03 29 03.760	+31 16 03.430	85.8	+1	4.0×3.6	8.7	-7.0 to 4.0	12.0-23.0
Per-emb 46		L1455	03 28 00.420	+30 08 01.010	228.6	+1	3.2×2.8	5.2	1.5-4.0	7.0-9.5
Per-emb 50		NGC 1333	03 29 07.764	+31 21 57.162	101.4	+1	1.8×1.5	7.6	1.5-4.0	11.0-13.5
Per-emb 52		NGC 1333	03 28 39.699	+31 17 31.882	335.9	+1	1.9×1.5	8.3	6.0-7.5	8.5-10.0
Per-emb 53	B5-IRS1		03 47 41.577	+32 51 43.745	275.4	+1	2.0×1.7	11.0	3.0-8.0	12.0-17.0
Per-emb 56	IRAS 03439+3233		03 47 05.422	+32 43 08.330	310.4	+1	2.2×2.1	11.0	6.0-10.0	12.0-16.0
Per-emb 57		NGC 1333	03 29 03.322	+31 23 14.338	202.7	+1	3.5×3.0		1.5-5.5	9.5-13.5

Table 1 – continued

Source	Common name	Region ^a	Protostar RA ^b (J2000)	Protostar Dec. ^b (J2000)	T_{bol}^c (K)	Robust	Beam major and minor axes (arcsec) ^d	v_{systemic} (km s ⁻¹) ^e	v_{blue} (km s ⁻¹)	v_{red} (km s ⁻¹)
Per-emb 61		IC348	03 44 21.301	+31 59 32.526	271.3	+1	2.0×1.7	9.3	4.0–7.5	10.5–14.0
Per-emb 62		IC348	03 44 12.973	+32 01 35.289	320.6	+1	2.9×2.5	8.6	1.0–6.5	11.5–17.0
B1-bS		B1	03 33 21.340	+31 07 26.440	14.7	+1	3.4×2.8		1.0–4.0	8.0–11.0
L1448N-NW	L1448N-IRS3C	L1448	03 25 35.673	+30 45 34.357	31.7	-1	1.4×1.2	5.3	0.0–3.0	7.0–10.0
L1448N-B	L1448N-IRS3B	L1448	03 25 35.330	+30 45 14.810	50.3	-1	1.4×1.2	5.3	0.5–3.0	7.0–9.5
Per-Bolo 45		NGC 1333	03 29 06.770	+31 17 29.960	15.0	+1	3.1×2.2		7.0–7.5	9.5–10.0
Per-Bolo 58		NGC 1333	03 29 25.417	+31 28 15.000	20.1	+1	3.1×2.2	8.2	3.0–6.5	9.5–13.0
SVS 13C		NGC 1333	03 29 02.030	+31 15 37.750	42.6	+1	3.9×3.5	8.9	4.0–7.0	9.0–12.0

^aThe clustered region of Perseus in which this protostar is located, if any. Protostars with listed regions can be found in the figures presented in Appendix A.

^bProtostellar positions traced by the MASSES 1.3 mm continuum detections when available (Pokhrel et al. 2018; Stephens et al. 2018), or by the VANDAM (Tobin et al. 2016, 2018).

^c T_{bol} values tabulated by Pokhrel et al. (2023).

^dAs compiled by Stephens et al. (2019).

^eAs measured by Stephens et al. (2018, 2019) using MASSES C¹⁸O (2 – 1) observations; missing values indicate MASSES C¹⁸O (2 – 1) non-detections.

500 pc) survey (Pokhrel et al. 2023),² the image weighting we used to make our measurements (see the following paragraph below), the major and minor axes of the MASSES ¹²CO (2 – 1) synthesized beam for each target using the image weighting listed in the previous column, the systemic velocity of each protostar when available from MASSES C¹⁸O (2 – 1) fits by Stephens et al. (2018, 2019), and the velocity intervals used for the fitting procedure described below in Section 3.

The MASSES survey produced and publicly delivered images with two different weightings, one with a robust parameter of +1 and one with a robust parameter of -1. While we refer to Stephens et al. (2019) for details, we note here that, compared to the robust = +1 images, the robust = -1 images have higher spatial resolution at the cost of reduced sensitivity and reduced recovery of large-scale emission. For the majority of the outflows identified here, we use the robust = +1 images to measure opening angles due to their improved sensitivity and improved recovery of large-scale emission. Eight poorly resolved outflows in the robust = +1 images are both better resolved and robustly detected in the robust = -1 images, thus for these eight outflows we use the robust = -1 images instead and indicate as such in the sixth column of Table 1. For outflows that are well detected and well resolved in both image weightings, our measured opening angles agree to within the uncertainties.

Notes on the detected outflows (and their driving sources) relevant for measuring their opening angles are given in Appendix B. Table 2 lists the source (and any associated common names), region (if applicable), phase centre of the MASSES observations, and T_{bol} of all MASSES targets for which we are unable to obtain outflow opening angle measurements. Relevant details on these targets are given in Appendix C, focusing on what is seen in the MASSES ¹²CO (2 – 1) observations and the reason(s) why we are unable to obtain opening angle measurements.

3 METHOD

3.1 Measuring outflow opening and position angles

Given the large number of outflows in our sample and the great diversity in observed outflow morphologies, we desire to measure outflow opening angles using a systematic, repeatable method rather than ‘by eye’ using a protractor. We also seek a method that can capture with a single angle the relative width of different outflows even if they do not exhibit simple, conical morphologies. To accomplish this, we adopt a method that was first developed by Offner et al. (2011) using synthetic (simulated) observations of protostellar outflows. This method accurately reproduces the opening angle that would be determined ‘by eye’ for a conical outflow, and returns the opening angle of the approximately conical base for curved outflow morphologies (see Offner et al. 2011, especially their fig. 2).

We begin by using our MASSES ¹²CO (2 – 1) observations to construct integrated intensity (moment 0) maps of each outflow. We manually select (by eye) the velocity intervals to use for each source, considering separately the redshifted and blueshifted velocities. We cut out the low-velocity channels close to the systemic cloud velocity, as these channels typically feature significant artefacts from the ambient cloud emission that is partially or fully resolved

²These updated values of T_{bol} were calculated by Pokhrel et al. (2023) using photometric and spectroscopic measurements in the 1 – 850 μm wavelength range from the Two Micron All-Sky Survey, the *Spitzer Space Telescope*, the *Herschel Space Observatory*, and the James Clerk Maxwell Telescope.

Table 2. MASSES targets without measured outflow opening angles.

Source	Common name	Region ^a	Phase centre RA ^b (J2000)	Phase centre Dec. ^b (J2000)	T_{bol} ^c (K)
Per-emb 4		NGC 1333	03 28 39.10	+31 06 01.80	37.4
Per-emb 8		IC348	03 44 43.62	+32 01 33.70	47.9
Per-emb 12	NGC 1333 IRAS 4A	NGC 1333	03 29 10.50	+31 13 31.00	32.5
Per-emb 14	NGC 1333 IRAS 4C	NGC 1333	03 29 13.52	+31 13 58.00	31.6
Per-emb 18	NGC 1333 IRAS 7	NGC 1333	03 29 10.99	+31 18 25.50	43.8
Per-emb 32		IC348	03 44 02.40	+32 02 04.90	80.5
Per-emb 35	NGC 1333 IRAS 1, IRAS 03255+3103	NGC 1333	03 28 37.09	+31 13 30.70	59.8
Per-emb 37		NGC 1333	03 29 18.27	+31 23 20.00	32.1
Per-emb 38			03 32 29.18	+31 02 40.90	149.4
Per-emb 39			03 33 13.78	+31 20 05.20	60.9
Per-emb 43			03 42 02.16	+31 48 02.10	192.4
Per-emb 45		B1	03 33 09.57	+31 05 31.20	223.6
Per-emb 47	IRAS 03254+3050	NGC 1333	03 28 34.50	+31 00 51.10	234.3
Per-emb 48	L1455-FIR2	L1455	03 27 38.23	+30 13 58.80	245.6
Per-emb 49		NGC 1333	03 29 12.94	+31 18 14.40	325.0
Per-emb 51		NGC 1333	03 28 34.53	+31 07 05.50	127.0
Per-emb 54	NGC 1333 IRAS 6	NGC 1333	03 29 01.57	+31 20 20.70	20.9
Per-emb 55	IRAS 03415+3152	IC348	03 44 43.62	+32 01 33.70	209.1
Per-emb 58		NGC 1333	03 28 58.44	+31 22 17.40	303.2
Per-emb 59			03 28 35.04	+30 20 09.90	341.0
Per-emb 60		NGC 1333	03 29 20.07	+31 24 07.50	363.0
Per-emb 63		NGC 1333	03 28 43.28	+31 17 33.00	482.9
Per-emb 64			03 33 12.85	+31 21 24.10	440.3
Per-emb 65		NGC 1333	03 28 56.31	+31 22 27.80	536.0
Per-emb 66		IC348	03 43 45.15	+32 03 58.60	480.0
B1-bN		B1	03 33 21.19	+31 07 40.60	10.2
L1448 IRS2E		L1448	03 25 25.66	+30 44 56.70	15.0
L1451-mm	L1451-MMS		03 25 10.21	+30 23 55.30	15.0
SVS 13B		NGC 1333	03 29 03.42	+31 15 57.72	20.0

^aThe clustered region of Perseus in which this target is located, if any. Protostars with listed regions can be found in the figures presented in Appendix A.

^bThe phase centre of the SMA MASSES observation of this target.

^c T_{bol} values tabulated by Pokhrel et al. (2023).

Sidelobe Emission?

out by our interferometric observations. We include the lowest possible velocity channels that are not affected by ambient cloud emission since many outflows are widest at the lowest velocities. The exact velocity intervals over which we integrate are listed in the last two columns of Table 1, and we note here that the upper bound in velocity has no discernible effect on our results since the integrated intensity is typically dominated by the lowest velocity channels, where the outflows are both the widest and the brightest.

For each pixel in these integrated intensity maps detected above 3σ (where σ is the rms noise in the integrated intensity maps, calculated using the IDL procedure SKY), we calculate the angle between that pixel and an initial guess of the outflow axis. When available, this initial guess is taken from Stephens et al. (2017); otherwise it is chosen from visual inspection of the data. We manually mask all pixels with detections above 3σ that are not part of the outflow, such as those arising from other, nearby outflows, or from sidelobe emission that is not fully removed in the cleaning process. We also mask all pixels within 4 arcsec of each protostar in order to avoid the effects of excessive beam smearing, but otherwise do not apply radial constraints to the pixels included in the fit. We then plot the resulting distribution of angles and fit a Gaussian to the distribution. Following Offner et al. (2011), we define the opening angle of the outflow as the full width at quarter-maximum of this best-fitting Gaussian. The angle at which the Gaussian peaks is taken as a refinement to the initial guess for the outflow position

angle. We refer to Offner et al. (2011) for a full discussion of the validity of this fitting method for measuring outflow opening angles.

One complication we must account for is that, with the relatively high spatial resolution of our interferometric observations, the outflows in our sample frequently feature resolved cavity walls and partially or fully empty cavities at small angles close to the outflow axis. One such example is shown in Fig. 1 for Per-emb 2. The partially empty cavities result in a double-peaked distribution of angles to detected pixels, with a dip where the Gaussian distribution should peak due to the lack of detections close to the outflow axis. For these cases, which make up the majority of our sample, we restrict the Gaussian fit to the wings, omitting from the fit the small angles close to the outflow axis where the distribution turns over and decreases. Exactly what is meant by ‘the wings’, and the implications for the resulting uncertainties in measured opening angles, are discussed in Section 3.2 below.

We are able to successfully measure opening angles for 46 outflows in the MASSES sample. We classify these outflows into four different groups based on our confidence in our results. For outflows like the Per-emb 2 outflow (Fig. 1), there is no ambiguity in identifying the presence of an outflow, no ambiguity in identifying the pixels contained within the outflow, and the Gaussian fit provides an excellent by-eye match to both the distribution of pixel angles and the morphology of the outflow in our integrated blueshifted and redshifted ^{12}CO (2 – 1) maps. We classify these objects into the

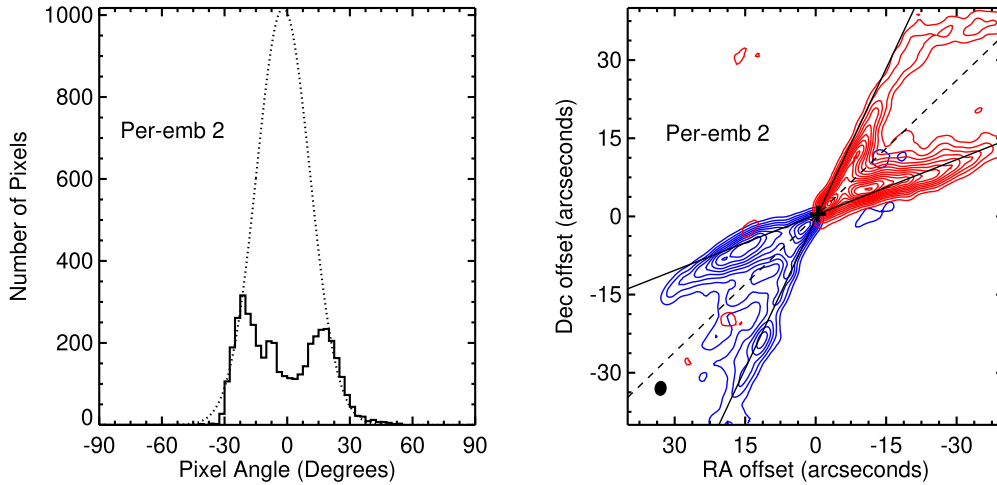


Figure 1. Left: the solid histogram shows the distribution of angles to all pixels in the outflow driven by Per-emb 2, as described in the text. The dotted line shows the best-fitting Gaussian distribution, where the fit was restricted to the wings in order to exclude the dip at low angles arising from the resolved, partially empty outflow cavity. Right: integrated blueshifted and redshifted ^{12}CO (2 – 1) emission showing the Per-emb 2 outflow, integrated over the velocity intervals reported in Table 5. The synthesized beam is shown as the filled ellipse in the lower left, and the cross marks the position of the protostar as traced by the 1.3 mm MASSES continuum detection (Pokhrel et al. 2018; Stephens et al. 2018). The dashed line shows the outflow position angle determined based on the peak angle of the Gaussian fit, and the solid lines show the opening angle determined from the full width at quarter-maximum of the Gaussian fit (see the text for details).

Table 3. Number of outflows in each group.

Group	Number
High confidence (HC)	22
Medium confidence (MC)	15
Low confidence (LC)	5
Upper limit	4

‘high confidence’ (HC) group. For other outflows, there may be an asymmetry between the two outflow lobes,³ or ambiguity in whether or not emission in some part of the map is part of the outflow, that require us to make subjective choices about what emission to include in the fit. Even though the quality of the fits are generally similar, the subjective choices add extra systematic uncertainties that are difficult to quantify, leading us to classify these outflows into the ‘medium confidence’ (MC) group. As a third category, some protostars drive outflows that, while clearly present, feature substantial ambiguity in their morphologies. We classify these outflows into the ‘low confidence’ group and emphasize that our measurements of them may be highly uncertain, with uncertainties in many cases that are difficult to quantify due to the subjective decisions in interpreting the complex ^{12}CO (2 – 1) maps. Finally, some outflows are partially or fully unresolved in one or both of their dimensions, and are thus classified as upper limits. Table 3 reports the numbers of outflows in each of these four categories.

3.2 Discussion of uncertainties

As described above, we restrict the majority of our Gaussian fits to the distributions of outflow pixel angles to the wings of the distributions.

³Whenever such asymmetries exist we restrict the fit to the lobe that appears visually wider, in an effort to capture the full opening angles of each outflow under the assumption that the asymmetries are due to partial non-detections of intrinsically symmetrical outflows. If the asymmetries are instead intrinsic the effective opening angle of the two lobes combined would be smaller in these cases than reported here.

These ‘wings’ are selected manually by specifying two different intervals over which to simultaneously fit (one interval at negative pixel angles and the other at positive pixel angles). The intervals are chosen such that they each start where the distributions first rise above zero, and they end just before the distributions turn over at angles close to the outflow axis. However, small adjustments to the bounds of these intervals do result in small changes to the best-fitting opening angles. We demonstrate this below in the left panel of Fig. 2, which shows three different best-fitting Gaussians for Per-emb 2 that result from three slightly different fitting intervals (the intervals themselves are specified in Table 4).

All three of the fits shown in the left panel of Fig. 2 adequately fit the data, but Fit2 gives an opening angle 7° smaller than Fit1, whereas Fit3 gives an opening angle 5° larger than Fit1. Given that the formal fit uncertainties are typically less than 1°, the choice of fitting interval is the dominant source of uncertainty in our results. To properly quantify this, we vary each of the four interval bounds, in steps of the bin size, by up to $\pm 20^\circ$ from our nominal, manually chosen bins. For each possible combination of the four bounds (except those where the negative or positive lower bounds exceed the corresponding upper bounds, which we remove from consideration), we perform the Gaussian fit, manually rejecting those fits that are obviously bad but keeping all others. The right panel of Fig. 2 shows the resulting distribution of best-fitting opening angles for the outflow driven by Per-emb 2, with the result obtained from the nominal, manually chosen interval indicated with a dotted line. The standard deviation of this distribution is 8°.

We repeat this process for each outflow in our sample. We then add in quadrature the standard deviation of the distribution of possible outflow opening angles, determined as described above, with the formal fit uncertainty for the fit performed over the nominal, manually chosen interval. We treat this resulting value as the uncertainty for each outflow opening angle. We do not apply this method for the LC outflows, and thus do not determine or list uncertainties for the LC outflow opening angle measurements, since the true uncertainties for these outflows are dominated by the subjective choices made in identifying the pixels containing outflow emission and are thus not easily quantifiable.

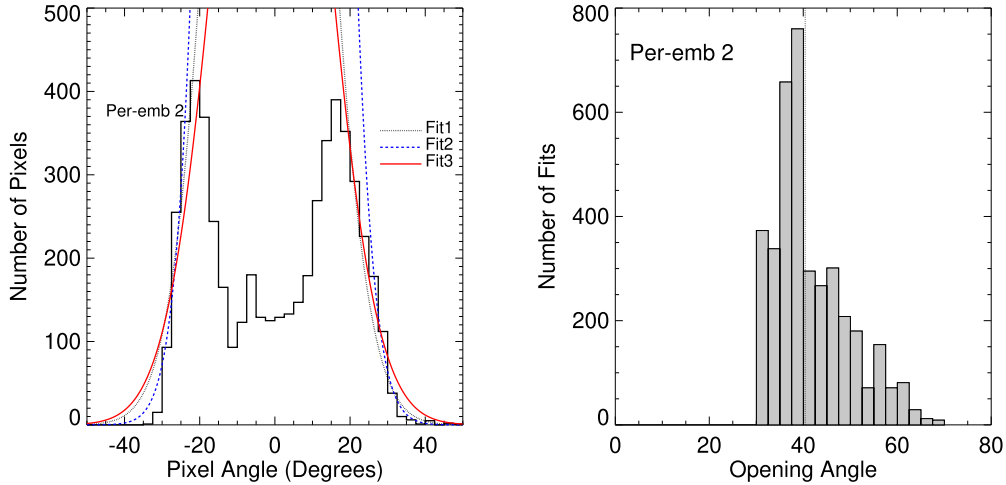


Figure 2. Left: the solid histogram shows the distribution of angles to all pixels in the outflow driven by Per-emb 2, as described in the text (this is the same distribution as that shown in Fig. 1). The three different dashed lines show three different best-fitting Gaussian distributions, with each of the three fits performed over slightly different intervals of pixel angles. The intervals used for each fit are listed in Table 4. Right: the distribution of all possible opening angles for the Per-emb 2 outflow, determined as described in the text. The dotted vertical line shows the opening angle determined from our fit to the nominal, manually selected interval over which to perform the Gaussian fit.

Table 4. Intervals for the fits shown in Fig. 2.

Fit	Negative lower bound (deg)	Negative upper bound (deg)	Positive lower bound (deg)	Positive upper bound (deg)
Fit1	−38.75	−23.75	18.75	38.75
Fit2	−48.75	−23.75	25.50	50.00
Fit3	−48.75	−17.00	25.50	50.00

While the full results are tabulated below, we note here that the uncertainties determined via this method range from 5–31°, but most typically range between 5–15°.

3.3 Validation of method

While our method for measuring opening angles was developed and validated by Offner et al. (2011), we further validate this method in two ways: (1) by comparing to outflow opening angles measured by hand from the same data, and (2) by applying our method to the outflow maps observed by Arce & Sargent (2006).

3.3.1 Comparison to measurements by hand

The results from applying our method to the MASSES data were compared to measurements of the same outflow opening angles made by the more traditional (and also more subjective) ‘by eye with a protractor’ method. A student with no prior experience with protostellar outflows attempted to independently identify outflows by manually inspecting the ^{12}CO (2 – 1) spectral cubes, collapsing the cubes into integrated intensity (moment 0) maps over the relevant velocity channels, and then measuring the opening angles of the outflows manually with a protractor. We note that the use of a student with no prior experience was intentional in order to apply a ‘sanity check’ on our automated method. Comparing to the results from this automated method, which will be presented and discussed in the following sections of this paper, we find that the mean difference between the automated and ‘by-eye’ methods is $13.3^\circ \pm 5.6^\circ$. Given

that the mean difference is within 3σ of 0° , we find that our automated method does generally agree with manual ‘by-eye’ measurements.

Digging more deeply into the results, we note that four outflows (Per-emb 27, Per-emb 30, Per-emb 44, and Per-emb 62) featured ‘by-eye’ measurements that disagree with the results from our automated method by greater than 20° . All four cases can be explained as errors resulting from the use of a student who did not possess background knowledge on these objects, including confusion due to multiple outflows in a binary system (Per-emb 27), unfamiliarity with the results from previously published, lower resolution maps that recover more extended emission (Per-emb 44), and improper selection of the velocity intervals for the integrated intensity (moment 0) maps (Per-emb 30 and Per-emb 62). When excluding these four outflows, the mean difference between these ‘by-eye’ measurements and the results of our automated method is $3.5^\circ \pm 2.1^\circ$. Thus, we find excellent agreement between the two methods, indicating that our method does pass this basic ‘sanity check’.

3.3.2 Comparison to Arce & Sargent (2006)

Arce & Sargent (2006) obtained interferometric observations of nine protostellar outflows and measured their opening angles manually using a protractor. While we have already demonstrated that our method provides visually good fits to the opening angles of our outflows, we further quantify this by comparing the results of applying our method to the Arce & Sargent (2006) data with their manual opening angle measurements.

Out of their nine observed outflows, Arce & Sargent (2006) were able to measure opening angles for seven of them. Five out of these seven would be classified into the HC group by our method. The remaining two (RNO 43 and T Tau) would be classified into either the MC or LC groups based on the presence of asymmetrical outflow lobes and substantial subjective uncertainties in what emission is and is not interpreted to be part of the outflow. Since the MC and LC groups include resulting systematic uncertainties that are difficult to fully quantify, we restrict our comparison to the five outflows that we would have placed in the HC group.

For all five of the HC outflows our method results in opening angles that agree with Arce & Sargent (2006) to within 15° , with four out of the five agreeing to within 10° . The mean difference between our measurements and those reported by Arce & Sargent (2006) is $4.2^\circ \pm 3.8^\circ$. Our measurements thus show strong statistical agreement with those of Arce & Sargent (2006), with a mean difference that is within 2σ of 0° . Furthermore, the dispersion in differences is comparable to many of our opening angle uncertainties that were determined as discussed in Section 3.2.

Finally, we note that there is one outflow in both our sample and the Arce & Sargent (2006) sample: Per-emb 5 (IRAS 03282 + 3035). Using our MASSSES data and our method, we measured an opening angle of 50° . Using the Arce & Sargent (2006) data and our method, we measured an opening angle 53° . Using their data and their manual method, Arce & Sargent (2006) measured an opening angle of 50° . Given the typical uncertainties of 5° – 10° reported in this paper, combined with the above analysis, we thus conclude that our method is in excellent agreement with methods based on manual, ‘by-eye’ measurements using a protractor.

4 RESULTS

Table 5 reports the positions and opening angles, along with the uncertainties in the opening angles and the confidence group assigned to each outflow, for the 46 outflows in the MASSSES sample for which we are able to obtain valid measurements. The T_{bol} values for each object are also repeated from Table 1 for convenience. Notes on each outflow for which we measure an opening angle, including justification for the confidence group assigned for that outflow, are presented in Appendix B. Additional notes on MASSSES pointings for which we are unable to measure outflow opening angles are given in Appendix C. We emphasize here that all opening angles reported in this paper are not corrected for inclination and are thus projected onto the plane of the sky. The effects of inclination corrections are discussed in Section 5.3.3.

Figs 3–5 plot blueshifted and redshifted integrated intensity (moment 0) contours for each of the 46 outflows, along with lines showing the position and opening angles measured for each outflow. Visual inspection of these figures reveals two key points about our method: (1) our method agrees very well with the cones that would be drawn by hand for conical outflows (examples include Per-emb 2, Per-emb 5, Per-emb 20, among others), in agreement with the validation results presented in Section 3.3, and (2) our method accurately measures the opening angles of the conical (or approximately conical) bases of outflows displaying more parabolic morphologies (examples include Per-emb 3, Per-emb 53, and Per-emb 62, among others).

Fig. 6 plots, for each of the 46 outflows in our sample, the measured opening angle versus the bolometric temperature (T_{bol}) of the driving source on both logarithmic and linear axes. Due to the large uncertainties with the measurements in the low confidence group, for the remainder of this paper we focus only on the results for the 37 outflows in the HC and MC groups.

Initial visual inspection of these figures suggests a significant difference between the opening angles of the outflows driven by the Classes 0 and I protostars. There appears to be an increasing trend in the Class 0 opening angles, with a distinct lack of the widest outflows. There is no clear visual evidence for a continuation of this trend in Class I, although there is a distinct lack of the most collimated outflows. To further illustrate this difference, panel (a) of Fig. 7 plots the distributions of the 23 Class 0 and 14 Class I opening angles (MC and HC only). The median opening angles are 49° for Class 0 and

Table 5. Position and opening angle fitting results.

Source	T_{bol}^a (K)	Position angle ($^\circ$)	Full opening angle ($^\circ$) ^b	Confidence group ^c
Per-emb 1	24.1	115	31 ± 5	HC
Per-emb 2	30.9	131	40 ± 8	HC
Per-emb 3	33.8	96	49 ± 11	HC
Per-emb 5	35.9	123	50 ± 8	HC
Per-emb 6	75.8	58	60 ± 5	MC
Per-emb 7	35.3	171^d	$< 25^d$	UL
Per-emb 9	37.9	56	31 ± 5	MC
Per-emb 10	34.6	49	41 ± 11	HC
Per-emb 11	34.4	162	29 ± 5	HC
Per-emb 13	25.0	172^d	$< 34^d$	UL
Per-emb 15	51.5	150	89 ± 5	HC
Per-emb 16	37.4	177	$91 \pm \dots$	LC
Per-emb 17	54.8	64	69 ± 8	HC
Per-emb 19	59.5	146	73 ± 13	MC
Per-emb 20	62.1	126	57 ± 14	HC
Per-emb 21	56.5	57^d	$46 \pm \dots^d$	LC
Per-emb 22	43.9	131	56 ± 7	HC
Per-emb 23	41.7	61	39 ± 12	HC
Per-emb 24	67.0	92	62 ± 5	HC
Per-emb 25	63.4	107	51 ± 11	HC
Per-emb 26	41.9	160^d	40 ± 6^d	HC
Per-emb 27	50.2	14^d	65 ± 11^d	MC
Per-emb 28	59.2	121^d	16 ± 5^d	MC
Per-emb 29	47.4	133	37 ± 7	MC
Per-emb 30	96.9	115	107 ± 9	HC
Per-emb 31	76.3	149^d	45 ± 13^d	MC
Per-emb 34	97.6	54	84 ± 9	MC
Per-emb 36	95.1	13^d	162 ± 31^d	MC
Per-emb 40	128.9	109^d	43 ± 8^d	MC
Per-emb 41	212.6	25	< 35	UL
Per-emb 42	48.8	50^d	49 ± 14^d	MC
Per-emb 44	85.8	140^d	144 ± 20^d	MC
Per-emb 46	228.6	133	79 ± 11	HC
Per-emb 50	101.4	95	55 ± 12	HC
Per-emb 52	335.9	12	51 ± 5	MC
Per-emb 53	275.4	60	90 ± 11	HC
Per-emb 56	310.4	145	63 ± 11	HC
Per-emb 57	202.7	148^d	$25 \pm \dots^d$	LC
Per-emb 61	271.3	10	60 ± 5	MC
Per-emb 62	320.6	18	73 ± 6	HC
B1-bS	14.7	110^d	$< 63^d$	UL
L1448N-NW	31.7	130^d	42 ± 8^d	HC
L1448N-B	50.3	129^d	54 ± 14^d	MC
Per-Bolo 45	15.0	142^d	$9 \pm \dots^d$	LC
Per-Bolo 58	20.1	87	23 ± 5	HC
SVS 13C	42.6	7^d	$67 \pm \dots^d$	LC

^aRepeated from Table 1 for ease of access.

^bNo uncertainties are presented for the measured opening angles in the low confidence group. See Section 3.2 for details.

^cHC: high confidence. MC: medium confidence. LC: low confidence. UL: upper limit.

^dBased on one outflow lobe (see Appendix B for details).

73° for Class I. A K-S test returns $p = 0.017$, indicating that the null hypothesis that the two distributions are drawn from the same parent distribution can be rejected with 98.3 per cent confidence. Thus, we conclude there is a statistically significant difference in the opening angles of Classes 0 and I outflows, with the Class I outflows wider than the Class 0 outflows.

We now examine trends between outflow opening angles and bolometric temperatures. For the Class 0 outflows, we calculate correlation coefficients of approximately 0.5 (0.53 in linear–linear

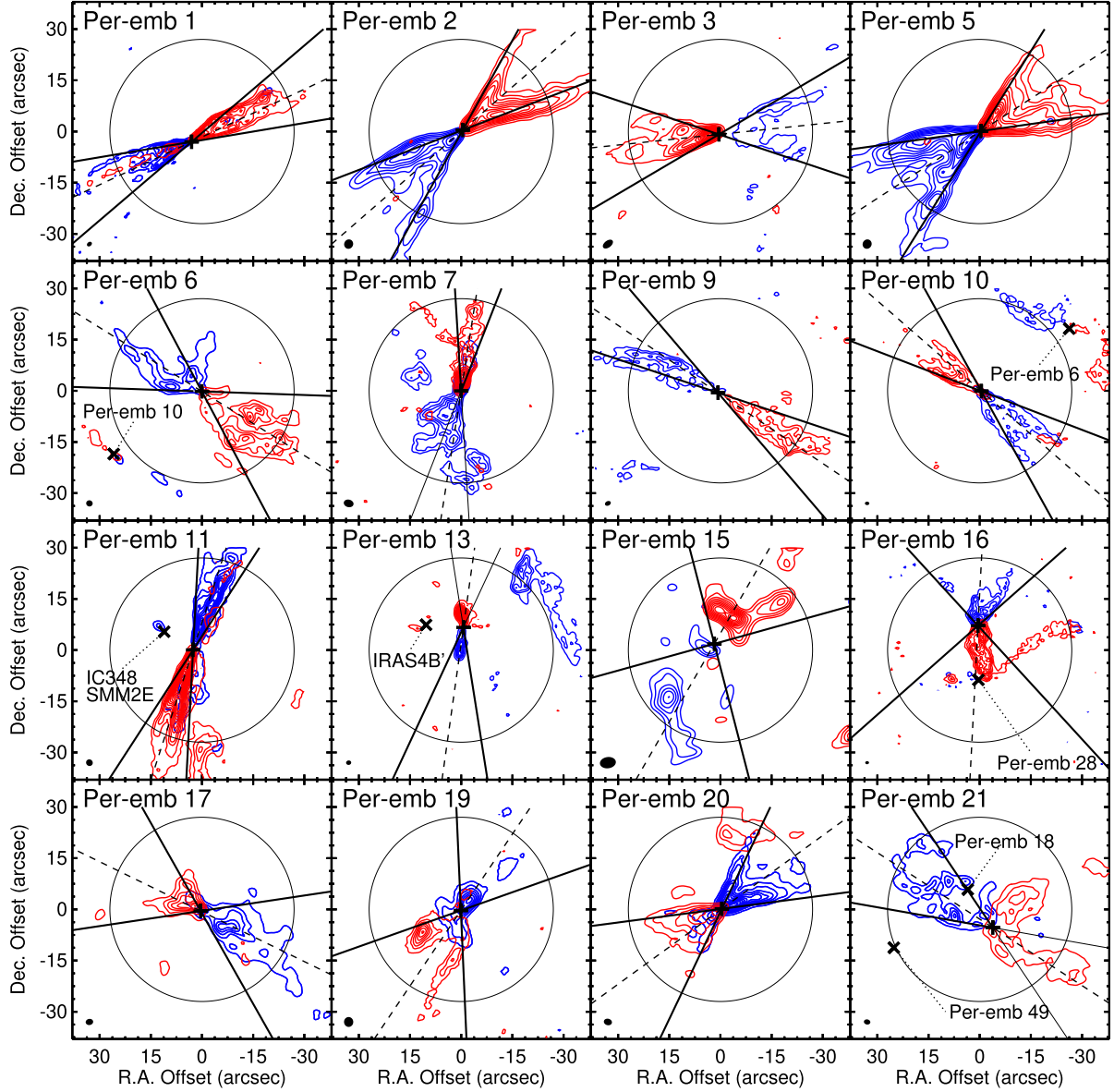


Figure 3. Blueshifted (blue contours) and redshifted (red contours) integrated intensity (moment 0) ^{12}CO (2 – 1) maps for Per-emb 1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 15, 16, 17, 19, 20, and 21. The velocity ranges over which the emission is integrated are given in Table 1. Driving source positions are marked with a cross, and the locations of other sources of interest in each field are marked with an x. The axes are plotted using coordinate offsets from the phase centres of the observations. The large circles show the FWHM size of the SMA primary beam at 230 GHz, centred on the phase centres. The cones plotted with thick solid lines show the opening angles of the outflows as measured in this work, with the position angles plotted using dashed lines. If an outflow lobe is not used in the fit, the corresponding cone is plotted with a thin, rather than a thick, solid line. The synthesized beams are shown in the bottom left corner of each panel. All emission contours and overplotted lines indicating outflow opening and position angles are masked at declination offsets greater than 30 arcsec to improve the clarity of the labelling of each panel. Contours start at 3σ and increase in steps of 4σ , where σ represents the 1σ rms in the blueshifted (for blue contours) or redshifted (for red contours) integrated intensity maps. The only exceptions are for Per-emb 7 and Per-emb 15, where the contours start at 2.5σ and increase in steps of 1σ .

space and 0.49 in log–log space), indicating a moderate positive linear correlation. For the Class I outflows, we calculate correlation coefficients of -0.30 in linear–linear space and -0.21 in log–log space, indicating either no linear correlation or a very weak negative linear correlation. Panel (b) of Fig. 7 displays the results of separate power-law fits (linear least-squares fits in log–log space) to the Classes 0 and I results. Defining θ to be the outflow opening angles in degrees, the best-fitting lines are:

$$\log(\theta) = (0.55 \pm 0.26) \log(T_{\text{bol}}) + (0.72 \pm 0.42), \quad \text{Class 0}, \quad (1)$$

$$\log(\theta) = (-0.05 \pm 0.12) \log(T_{\text{bol}}) + (1.92 \pm 0.27), \quad \text{Class I} \quad (2)$$

A Class 0 slope that is 2.1σ above zero and a Class I slope that is below zero by a statistically insignificant amount (0.4σ) support our above statements that the Class 0 outflows have opening angles that exhibit a moderate positive linear correlation with T_{bol} , whereas the Class I outflows have opening angles that exhibit either no, or a very weak negative, linear correlation with T_{bol} .

To better examine where the transition occurs between an increasing trend of outflow opening angle with T_{bol} and no trend,

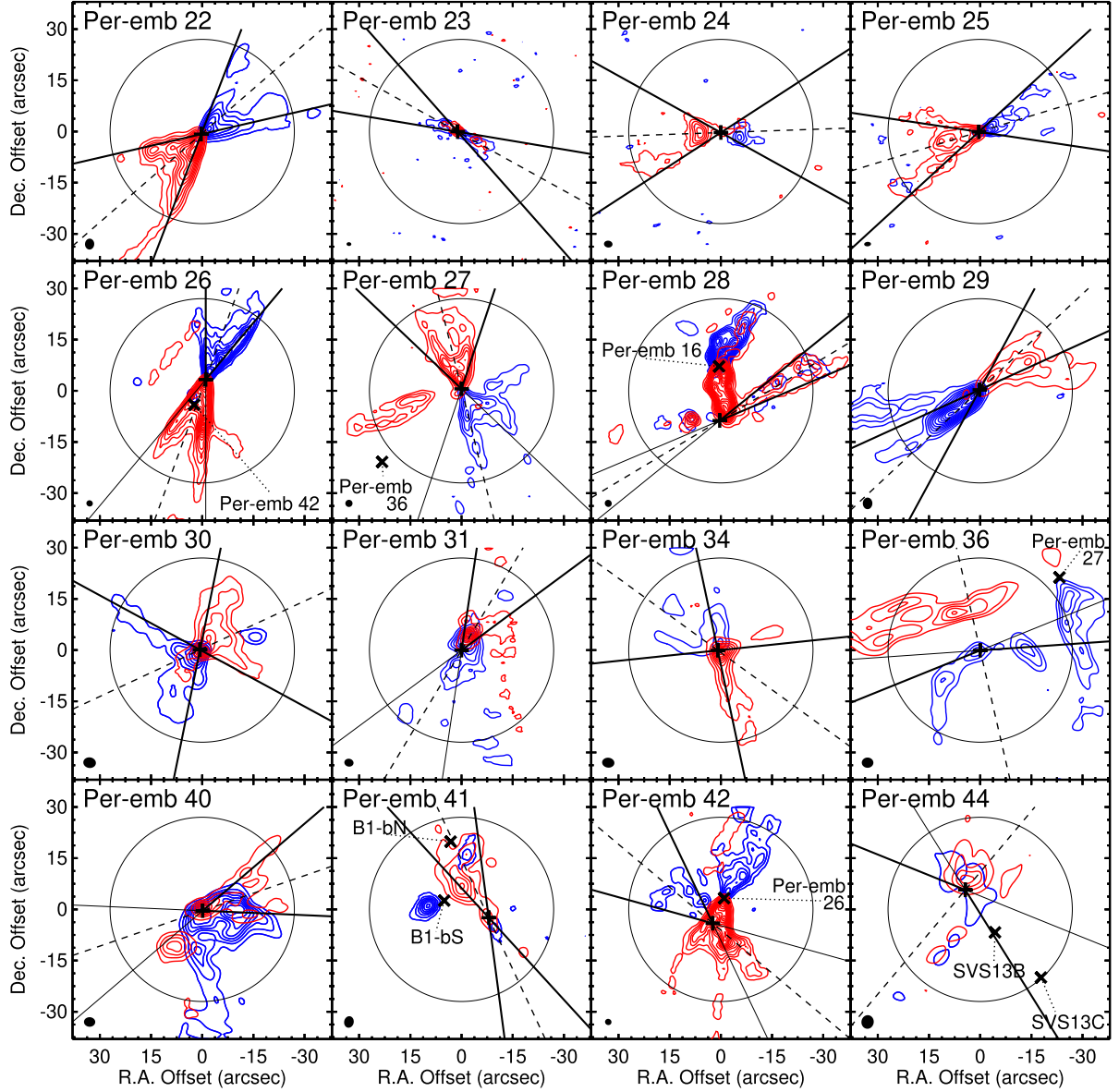


Figure 4. Same as Fig. 3, except for Per-emb 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 34, 36, 40, 41, 42, and 44. Contours start at 3σ and increase in steps of 4σ , where σ represents the 1σ rms in the blueshifted (for blue contours) or redshifted (for red contours) integrated intensity maps. The only exception is for Per-emb 36, where the contours start at 2.5σ and increase in steps of 2.5σ .

we repeated the linear least-squares fits in log–log space with the break point treated as an additional free parameter rather than holding it fixed at $T_{\text{bol}} = 70$ K. Both the parameters of the fits and the quality of the fits were statistically consistent over a broad range of T_{bol} values for the break point, between approximately $40 \text{ K} < T_{\text{bol}} < 100 \text{ K}$, lending support to the notion that there is a transition in the growth of outflow opening angles near the Class 0/I boundary.

Continuing with this analysis, we next tried fitting the following two functions, both of which increase at low values of T_{bol} and asymptote to constant values at high values of T_{bol} :

$$\theta = \theta_0 \tanh \left(\frac{T_{\text{bol}} T_{\text{bol,shift}}}{T_{\text{bol},0}} \right) \quad (3)$$

and

$$\theta = \theta_0 \left(1 - e^{-\frac{(T_{\text{bol}} - T_{\text{bol,shift}})}{T_{\text{bol},0}}} \right). \quad (4)$$

In both fits the parameter θ_0 is the opening angle that the functions asymptote to at large values of T_{bol} , the parameter $T_{\text{bol,shift}}$ sets the horizontal shift of the functions, and the parameter $T_{\text{bol},0}$ controls how quickly the functions transition from increasing to approximately constant. We emphasize here that our fits are not meant to actually describe or determine the underlying mathematical relationship between outflow opening angles and T_{bol} ; they are simply example functions that have turnover from increasing with T_{bol} to constant in T_{bol} , allowing us to explore when in the evolution of a protostar this transition occurs (see e.g. Heimsoth et al. 2022, for a similar fitting strategy to study the evolution of protostellar properties).

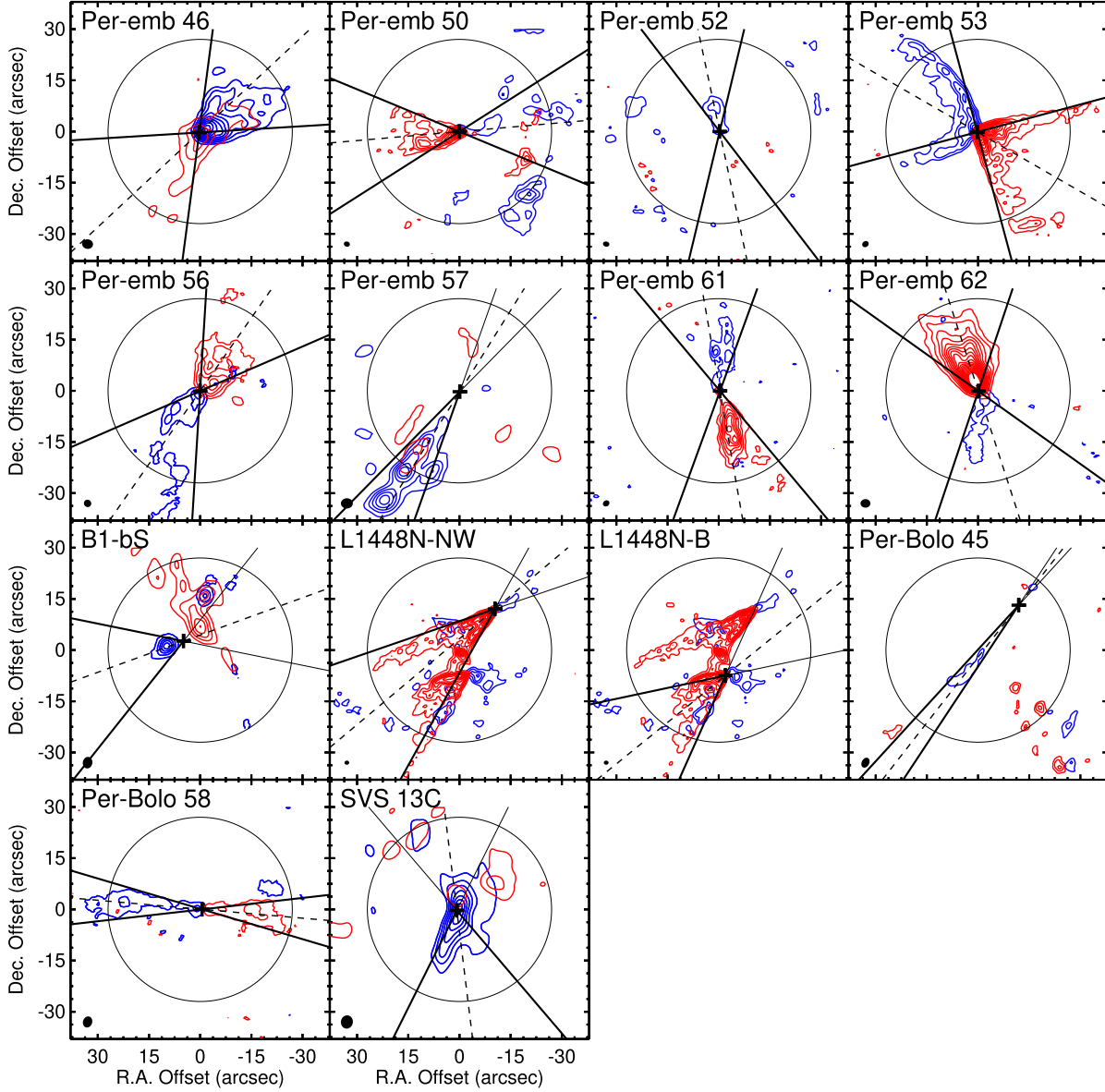


Figure 5. Same as Fig. 3, except for Per-emb 46, 50, 52, 53, 56, 57, 61, 62, B1-bS, L1448N-NW, L1448N-B, Per-Bolo 45, Per-Bolo 58, and SVS 13C. Contours start at 3σ and increase in steps of 4σ , where σ represents the 1σ rms in the blueshifted (for blue contours) or redshifted (for red contours) integrated intensity maps.

The best-fitting parameters are tabulated in the first two rows of Table 6, and the best-fitting functions are overplotted on our measurements in panel (c) of Fig. 7. The two fits show remarkable consistency with each other: they both asymptote to opening angles of 65° , and they reach 90 per cent of these maximum projected opening angles at T_{bol} values of 67 and 70 K for the hyperbolic tangent and $(1 - e)$ fits, respectively. Thus, for both functions nearly all of the increase in opening angle occurs during the Class 0 stage.

Since our error bars likely do not represent true statistical uncertainties (see Section 3.2), and thus may not be appropriate to use in weighted fits, we repeat the hyperbolic tangent and $(1 - e)$ fits with uniform weighting of each measurement, rather than weighting by the uncertainties. The results of these fits are tabulated in the third and fourth rows of Table 6, and the best-fitting functions are overplotted on our measurements in panel (d) of Fig. 7. The unweighted functions asymptote to somewhat larger opening angles

[78° for both the hyperbolic tangent and $(1 - e)$ fits], and they reach 90 per cent of their maximum projected opening angles at larger values of T_{bol} [74 and 79 K for the hyperbolic tangent and $(1 - e)$ fits, respectively]. However, as these T_{bol} values are still very close to the Class 0/I boundary of 70 K, our unweighted fits agree with the general conclusion from the weighted fits that nearly all of the increase in opening angle occurs in Class 0.

5 DISCUSSION

5.1 Comparison to previous results

Several studies have investigated the widths of outflow opening angles since the seminal work by Arce & Sargent (2006). Here, we compare our results to those obtained from these studies, focusing on studies that have used large samples for statistical analyses. Cor-

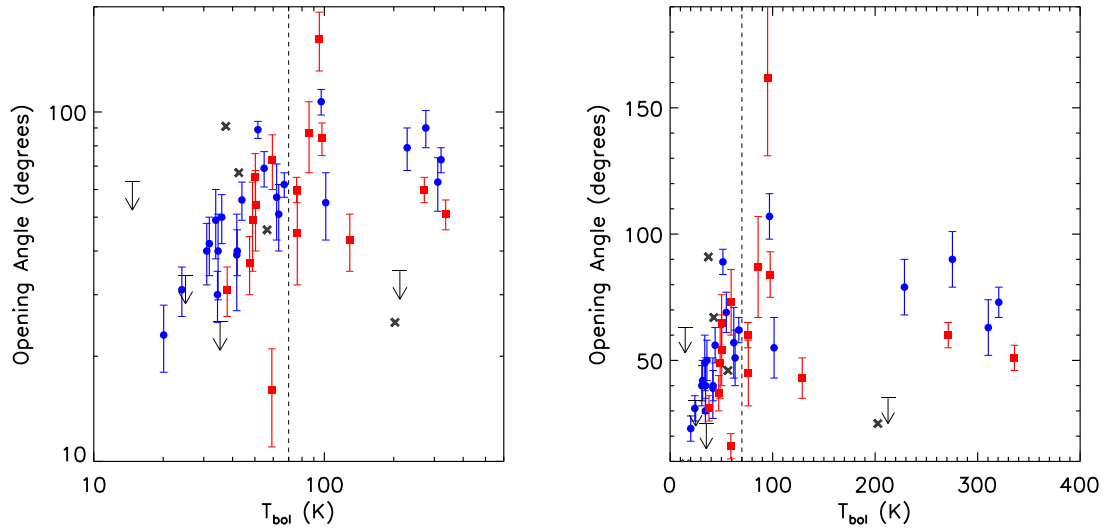


Figure 6. Measured outflow opening angles plotted versus the bolometric temperatures of the driving sources, on both logarithmic (left) and linear (right) axes. In both panels, the blue circles show HC measurements, the red squares show MC measurements, the black x's show low confidence measurements, and the arrows show upper limits. The dashed vertical line in each panel marks $T_{\text{bol}} = 70$ K, the Class 0/I boundary in bolometric temperature (Chen et al. 1995), with Class 0 protostars to the left ($T_{\text{bol}} < 70$ K) and Class I protostars to the right ($T_{\text{bol}} \geq 70$ K).

rections for inclination and methodology used to measure opening angles that will be necessary when comparing results from different studies are described in Appendix D, and will be referenced as necessary throughout this section.

5.1.1 Comparison to Seale & Looney (2008)

Seale & Looney (2008) measured the opening angles of 27 outflows using mid-infrared images from the *Spitzer Space Telescope* (approximately 2 arcsec resolution) that detect scattered light off the outflow cavity walls. They plotted angular intensity profiles at the largest radius from each protostar where the cavity walls were clearly detected, and then measured the angular distance between the two peaks corresponding to each of the cavity walls. While this method will give different results for parabolic outflows than our method, which measures the opening angle at the base of the outflow, the majority of their outflows appear conical, and indeed they noted that their results are relatively insensitive to the exact radii at which the measurements are made. Seale & Looney (2008) used their measurements to conclude that outflows do widen with age, with a distinct lack of the widest outflows at the youngest ages and a distinct lack of the most collimated outflows at the oldest ages (see in particular their fig. 4). While they did not find any evidence for a transition between widening and constant opening angles, their sample is dominated by younger objects (Classes 0 or 0/I transition objects) and thus such a transition would not be apparent in their data. Thus, at least in terms of general trends, their findings are consistent with our results.

Direct comparisons between their results and ours are hindered by the fact that they do not measure or report T_{bol} values for their driving sources, along with the fact that their classifications are based on two different luminosity ratios ($L_{\text{bol}}/L_{1.3}$ and $L_{\text{submm}}/L_{\text{bol}}$) that are calculated using incompletely sampled SEDs (especially in the far-infrared and submillimetre; see Dunham et al. 2008, 2013; Enoch et al. 2009, for detailed discussions about the reliability of luminosity calculations using such SEDs). Nevertheless, using their classifications, the median outflow opening angle is 46° for the nine outflows in the Seale & Looney (2008) sample that are classified as

Class 0 by both of their luminosity ratios (nearly identical to our Class 0 median opening angle of 49°). Furthermore, the median opening angle for the four outflows in the Seale & Looney (2008) sample that are classified as either Class 0/I or Class I by both of their luminosity ratios is 54° , supporting the conclusion that the median opening angle is narrower for the youngest sources (a quantitative comparison to our median Class I opening angle of 73° is not possible since their sample is dominated by younger objects).

Finally, Table 7 compares the MASSES and Seale & Looney (2008) opening angles for the eight outflows common to both samples. While the agreement appears poor, four outflows (Per-emb 2, Per-emb 17, Per-emb 30, and Per-emb 34) are very weakly detected by Seale & Looney (2008) in their mid-infrared scattered light images, rendering the reliability of this comparison too low for further analysis. We thus conclude that the general trends are in agreement between the two samples, with a more detailed quantitative comparison simply not possible.

5.1.2 Comparison to Velusamy et al. (2014)

Velusamy et al. (2014) measured the projected opening angles of 31 outflows using mid-infrared images from the *Spitzer Space Telescope* processed using the HiRes deconvolution algorithm (Backus et al. 2005) in order to achieve approximately 1 arcsec resolution (native *Spitzer* resolution is approximately 2 arcsec). They measured the opening angles of the scattered light cavities by hand, and they explicitly noted that they measured the opening angle at the base for parabolic outflows. Since our method agrees with opening angles measured by hand in this manner (see Offner et al. 2011, in particular their fig. 2), and also measures projected opening angles, our results can be directly compared without any conversions or adjustments.

The top left panel of Fig. 8 plots the outflow opening angles versus the T_{bol} of the driving sources for both our measurements and those of Velusamy et al. (2014). The median opening angles in the Velusamy et al. (2014) sample are 68° for Class 0 and 105° for Class I, as reported in Table 8, thus like the MASSES sample the Velusamy et al. (2014) sample shows an increasing trend between Classes 0 and I. Overplotted in Fig. 8 are unweighted hyperbolic

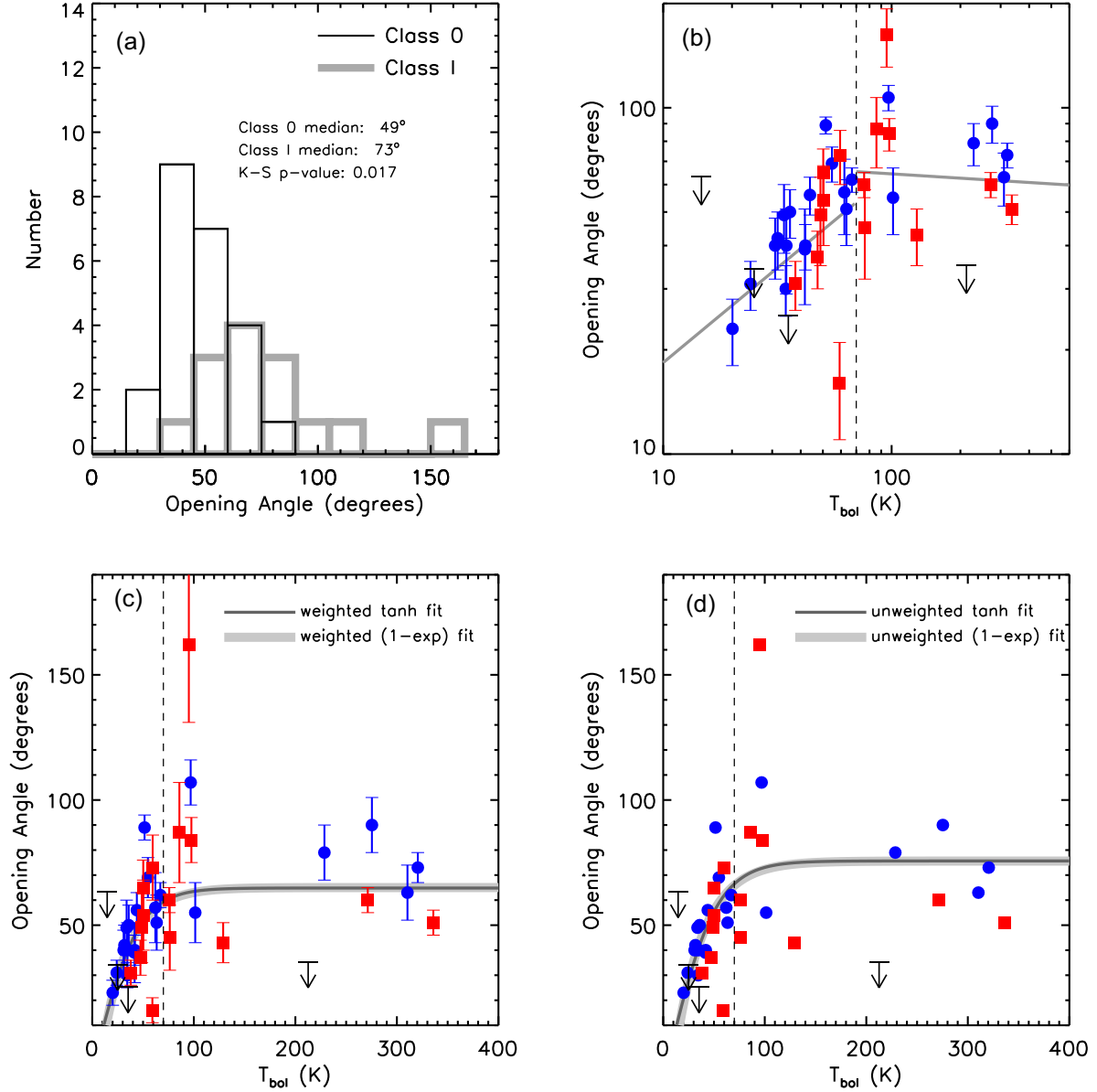


Figure 7. (a) Histograms showing the distributions of our measured Class 0 (thin black lines) and Class I (thick grey lines) outflow opening angles for the MC and HC groups. (b) Measured outflow opening angles versus T_{bol} plotted on logarithmic axes. The best-fitting two-component power law, with the break point fixed at $T_{\text{bol}} = 70$ K and the fit weighted by the error bars on each measurement, is overplotted with the thick grey line. (c) Measured outflow opening angles versus T_{bol} plotted on linear axes. The best-fitting hyperbolic tangent and exponential functions are overplotted with thin dark grey and thick light grey lines, respectively (see the text for details). The fits are weighted by the error bars on each measurement. (d) Same as panel (c), except the fits are unweighted. The error bars are not plotted to further emphasize that these are unweighted fits. In panels (b)–(d), MC measurements (red squares), HC measurements (blue circles), and upper limits (downward arrows) are all plotted but only the MC and HC opening angles are used in the fits.

Table 6. Best-fitting parameters for hyperbolic tangent and exponential fits.

Function	θ_0 (deg)	$T_{\text{bol,shift}}$ (K)	$T_{\text{bol},0}$ (K)
Weighted $\tanh$ (equation 3)	65	5	42
Weighted $(1 - e)$ (equation 4)	65	10	26
Unweighted $\tanh$ (equation 3)	78	9	44
Unweighted $(1 - e)$ (equation 4)	78	14	28

tangent fits to just the MASSES opening angles, to the combination of the MASSES and Velusamy et al. (2014) opening angles, and just the Velusamy et al. (2014) opening angles. The parameters of these fits are reported in Table 8. The qualitative agreement between our

Table 7. Common Sources in MASSES and Seale & Looney (2008).

Driving source	MASSES Opening angle (°)	Seale & Looney (2008) Opening angle (°)
Per-emb 2	40 ± 8	82 ± 8
Per-emb 17	69 ± 8	24 ± 8
Per-emb 25	51 ± 11	66 ± 8
Per-emb 26	40 ± 6	46 ± 4
Per-emb 30	107 ± 9	49 ± 8
Per-emb 31	45 ± 13	82 ± 4
Per-emb 34	84 ± 9	49 ± 6
Per-emb 40	43 ± 8	57 ± 8

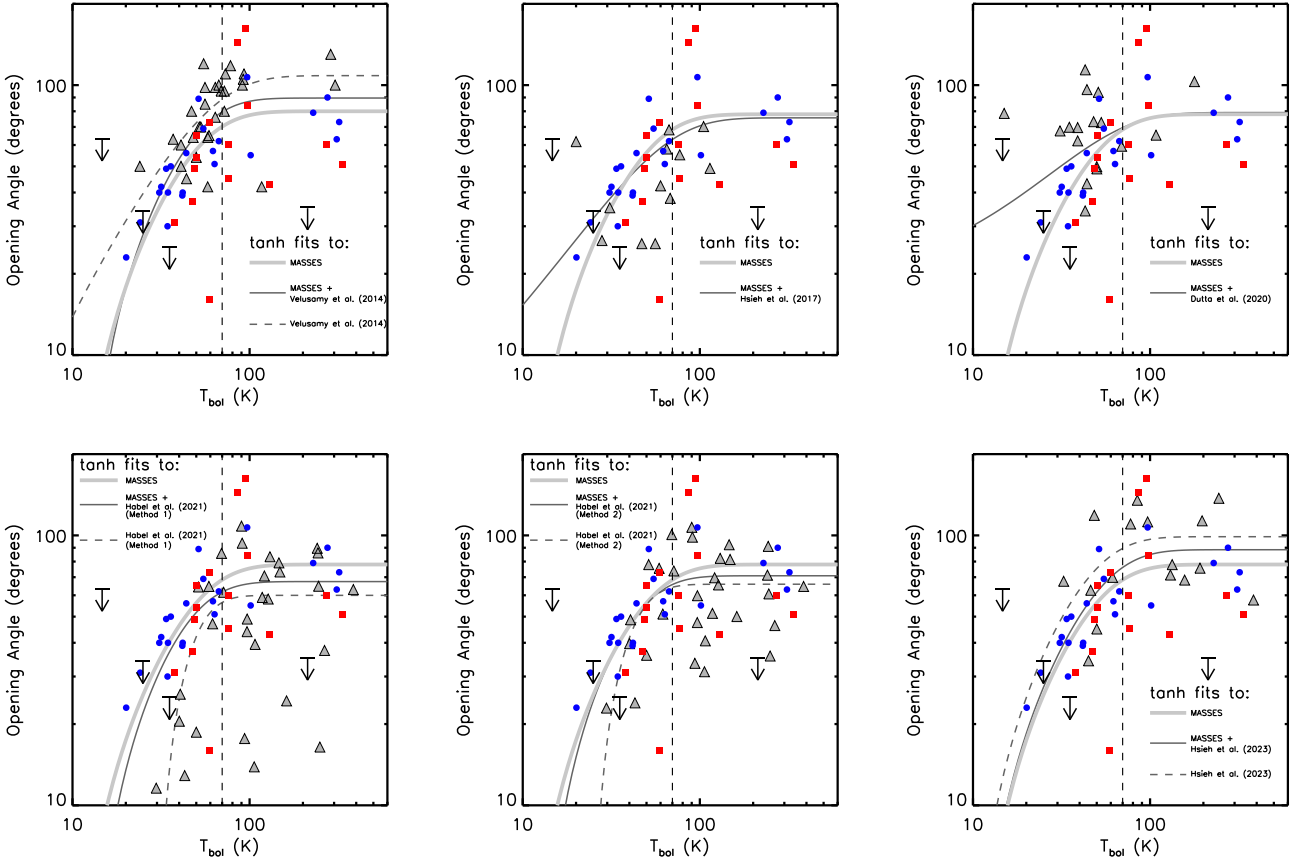


Figure 8. Outflow opening angles (projected onto the plane of the sky) plotted versus the bolometric temperatures of their driving sources, on logarithmic axes. In all panels, the blue circles plot the MASSES HC measurements from this work, the red squares plot the MASSES MC measurements from this work, the downward arrows plot the MASSES upper limits from this work, and the grey triangles plot the measurements from the previous study noted in the text in each panel. The dashed vertical line in each panel marks $T_{\text{bol}} = 70$ K, the Class 0/I boundary in bolometric temperature (Chen et al. 1995), with Class 0 protostars to the left ($T_{\text{bol}} < 70$ K), and Class I protostars to the right ($T_{\text{bol}} \geq 70$ K). The thick grey line in each panel shows the same unweighted hyperbolic tangent fit to the MASSES measurements that is shown in panel (d) of Fig. 7. The other lines in each panel show other unweighted hyperbolic tangent fits as noted in the text in each panel, and as described in the text in Section 5.1. Upper limits are not used in any of the fits presented here.

Table 8. Comparison to previously published surveys of outflow opening angles.

Data set	Median projected opening angle ^a		Unweighted hyperbolic tangent fits		
	Class 0 (deg)	Class I (deg)	θ_0 (deg)	$T_{\text{bol,shift}}$ (K)	$T_{\text{bol,0}}$ (K)
MASSES	49	73	78	5	42
Velusamy et al. (2014)	68	105	108	2	59
MASSES + Velusamy et al. (2014)	... ^b	... ^b	90	12	41
Hsieh et al. (2017)	38	55	... ^c	... ^c	... ^c
MASSES + Hsieh et al. (2017)	... ^b	... ^b	76	−2	61
Dutta et al. (2020)	70	84	... ^c	... ^c	... ^c
MASSES + Dutta et al. (2020)	... ^b	... ^b	79	−15	62
Habel et al. (2021) (Method 1)	26	62	60	30	23
MASSES + Habel et al. (2021) (Method 1)	... ^b	... ^b	67	13	37
Habel et al. (2021) (Method 2)	49	65	66	24	21
MASSES + Habel et al. (2021) (Method 2)	... ^b	... ^b	71	12	35
Hsieh et al. (2023)	65	110	99	10	40
MASSES + Hsieh et al. (2023)	... ^b	... ^b	88	10	46
MASSES + literature (Habel method 1)	57	76	79	9	38
MASSES + literature (Habel method 2)	57	76	81	9	39
MASSES + literature (no Habel)	58	90	90	4	51

^a All opening angles are full opening angles, and are projected onto the plane of the sky and thus uncorrected for inclination.

^b Median opening angles not calculated.

^c No fit performed due to a lack of Class I objects.

results is very good, and indeed Velusamy et al. (2014) also concluded that there is a widening trend for younger outflows that flattens considerably for more evolved outflows (although like our sample, theirs also contains about two times more Class 0 outflows than Class I outflows). While they did not explicitly attempt to constrain the location of this transition, they did present power-law fits (linear fits in logarithmic space) that assumed this transition occurs very close to the Class 0/I boundary. Thus, we find good agreement between their conclusions and the results from our this study.

Inspection of Fig. 8 suggests there may be a systematic offset to higher opening angles in the Velusamy et al. (2014) measurements compared to our own measurements, most notable when comparing the medians (68° versus 49° for Class 0; 105° versus 73° for Class I) and in the offset between the unweighted hyperbolic tangent fits solely to the MASSES results and solely to the Velusamy et al. (2014) results. Although this offset could be due to the different samples, there is some evidence that it is in fact due to the different outflow tracers used (mid-infrared scattered light versus ^{12}CO emission). Of the four outflows contained in both samples, the opening angles reported by Velusamy et al. (2014) are 30° wider (Per-emb 5), 4° wider (Per-emb 16), 19° wider (Per-emb 1), and 40° wider (Per-emb 53), giving mean and median differences of 23.25° and 24.5° , respectively. Velusamy et al. (2014) also noted this offset based on comparing to previously published ^{12}CO maps and suggested that opacity effects in the lowest velocity ^{12}CO channels where outflows are typically the widest may limit the ability of ^{12}CO to probe the full extent of outflow opening angles. Indeed, Arce et al. (2013) and Zhang et al. (2016) present a specific instance where a narrower opening angle is seen in ^{12}CO emission compared to both ^{13}CO emission and mid-infrared scattered light (HH 46/47; see in particular fig. 8 of Arce et al. 2013 and fig. 16 of Zhang et al. 2016). Future work is needed to more explicitly test the true extent to which opening angles measured in ^{12}CO underestimate the intrinsic opening angles due to both opacity and/or chemistry (e.g. Tychoniec et al. 2021) effects, using larger samples of outflows whose opening angles are measured using a variety of different tracers.

5.1.3 Comparison to Hsieh et al. (2017)

Hsieh et al. (2017) measured the opening angles of 12 outflows driven by low-luminosity protostars using near- and mid-infrared images with resolutions of approximately 1–2 arcsec, nine of which are Class 0 outflows and three of which are Class I outflows. To measure the opening angles they used the two-dimensional Whitney et al. (2003) radiative transfer models that include curved outflow cavities (specifically, outflow cavities following power laws with indices of 1.5) fit to the scattered light infrared images. They adopted the ‘Class Late 0’ model from Whitney et al. (2003) for all of their objects, with the opening angle and inclination of the outflow treated as free parameters and their reported opening angles corrected for inclination. Since their models adopt curved rather than conical outflow cavities, their opening angles are defined to be the opening angles of cones that reach the same width at a vertical distance of 5000 au above the mid-plane (see Appendix D).

Despite the significant methodology differences we can still perform a direct comparison between our results because the two methods give opening angles that are consistent within the typical uncertainties of 5° – 15° , as shown in the appendix of Myers, Dunham & Stephens (2023). Since our measurements are not corrected for inclination, in order for this comparison to be valid we take the opening angles reported by Hsieh et al. (2017) and project them back onto the plane of the sky, using equation (D4) with their reported

source inclinations. The top middle panel of Fig. 8 plots the outflow opening angles versus the T_{bol} of the driving sources for both our measurements and those of Hsieh et al. (2017, after projection back onto the plane of the sky). The median projected opening angles in the Hsieh et al. (2017) sample are 38° for Class 0 and 55° for Class I, as reported in Table 8. Overplotted in Fig. 8 are unweighted hyperbolic tangent fits to just the MASSES opening angles and to the combination of the MASSES and Hsieh et al. (2017) opening angles (we do not fit to just the Hsieh et al. (2017) opening angles since their sample contains so few Class I objects). The parameters of these fits are reported in Table 8.

Based on their measurements, Hsieh et al. (2017) concluded that there is a trend of increasing opening angle with T_{bol} . They do not find any clear evidence for a flattening of this trend, but their sample only contains three Class I outflows, all of which are close to the Class 0/I boundary. When considering their results combined with ours, the two samples given generally consistent results.

5.1.4 Comparison to Dutta et al. (2020)

Dutta et al. (2020) measured the outflow lobe widths and bolometric temperatures of the driving sources for 17 protostars in Orion originally identified as Orion Planck Galactic Cold Clumps. They measured the outflow lobe widths at distances of both 400 and 800 au from the driving source using subarcsecond resolution ^{12}CO ($2-1$) images obtained with the Atacama Large Millimeter/submillimeter Array (ALMA). We calculated opening angles at 800 au distances from the driving sources, using their measured widths at this distance. Dutta et al. (2020) did not correct for inclination, so these calculated opening angles can be directly compared to our opening angles, which are also uncorrected for inclination.

The top right panel of Fig. 8 plots the outflow opening angles versus the T_{bol} of the driving sources for both our MASSES measurements and those calculated by us using the Dutta et al. (2020) measurements. Overplotted are unweighted hyperbolic tangent fits to just the MASSES opening angles and to the combination of the MASSES and Dutta et al. (2020) opening angles (we do not fit to just the Dutta et al. 2020, opening angles since their sample contains so few Class I objects). The parameters of these fits are reported in Table 8. Also reported in Table 8 are the median Class 0 and Class I opening angles in the Dutta et al. (2020) sample.

Examining the Dutta et al. (2020) results alone does not reveal strong existence of a trend, consistent with their findings of at best a very weak positive correlation between outflow widths and T_{bol} . However, as they noted, the majority of their sample spans a very narrow evolutionary range in the latter part of the Class 0 evolutionary stage, with only two Class I objects and only one Class 0 object with $T_{\text{bol}} < 30$ K (this one young Class 0 object appears to be a significant outlier in that it has a large opening angle, and this outlier will be discussed further in Section 5.1.7 below). Fitting to the combined MASSES and Dutta et al. (2020) samples yields a hyperbolic tangent fit that asymptotes to 79° in the Class I stage, nearly identical to our value of 78° for the MASSES sample alone. Thus, while the Dutta et al. (2020) do not by themselves clearly show an increasing trend in Class 0 that transitions to a constant maximum projected opening angle in Class I, their results are not inconsistent with this result.

5.1.5 Comparison to Habel et al. (2021)

Habel et al. (2021) measured the opening angles of 30 outflows using 0.18 arcsec resolution near-infrared images from the *Hubble*

Space Telescope, 10 of which are Class 0 outflows and 20 of which are Class I outflows. To measure the opening angles they applied an edge detection algorithm to their near-infrared images in order to detect the edges of the cavity walls seen in scattered light. They then fit power laws to the cavity walls, treating the power-law index as a free parameter and finding mean and median indices of 1.9 and 1.5, respectively (indices of 1 and 2 indicate conical and parabolic morphologies, respectively). Since most of their outflows are not found to be conical, their opening angles are defined to be the opening angles of cones that reach the same width as the power-law cavities at a vertical distance of 8000 au above the mid-plane. Their reported opening angles are the half-angles of these cones, after applying model-derived correction factors to correct for inclination and other factors, as described below.

To perform an ‘apples-to-apples’ comparison between our MASSSES results, which are observed opening angles without inclination or other corrections, and those of Habel et al. (2021), we must ‘undo’ their corrections. Before attempting to undo their full model-derived corrections, we first explore the effects of inclination corrections alone. We first recalculate their opening angles to be those of cones that reach the same width at a vertical height of 5000 au above the mid-plane, rather than 8000 au above the mid-plane, using equations (D7)–(D9). This correction to a lower vertical height is chosen to be more consistent with both Hsieh et al. (2017) and with the comparison of methods presented in the appendix of Myers et al. (2023). Next, we multiply the opening angles reported by Habel et al. (2021) by a factor of two to convert from half opening angles to full opening angles. Finally, we project these corrected opening angles onto the plane of the sky, using equation (D4) with the source inclinations reported by Furlan et al. (2016, these inclinations are derived from SED modelling; see Furlan et al. for details on the modelling process and associated uncertainties). We refer to these correction steps as correction method 1.

Habel et al. (2021) applied model-derived correction factors to their measured cavity widths that are much larger than those derived from inclination corrections alone. They attribute these correction factors to the combined effects of inclination, the penetration of light into and beyond the cavity edges, and other systematic biases in their measurement technique, all of which caused their measured outflow opening angles to be wider than the intrinsic opening angles. Since both the MASSSES and all other literature measurements considered in Section 5.1 are direct measurements in the plane of the sky, without any correction for the various methodological biases present in each study, we ‘undo’ their model-derived corrections to obtain their directly measured opening angles. In practice, we do this by multiplying x_{8000} (see Fig. D2) by the Habel et al. model-derived correction factor for each target in their sample,⁴ recalculating their α_{8000} (again, see Fig. D2) using this larger width, converting from α_{8000} to α_{5000} following Appendix D, and then finally multiplying by a factor of two to convert from half to full opening angles. We refer to these correction steps as correction method 2.

The bottom left and centre panels of Fig. 8 plot the projected outflow opening angles versus the T_{bol} of the driving sources for both our MASSSES measurements and those presented by Habel et al.

(2021) after applying correction method 1 (left panel) or correction method 2 (right panel). Overplotted are unweighted hyperbolic tangent fits to the MASSSES opening angles, to the Habel et al. (2021) opening angles, and to the combination of the MASSSES and Habel et al. (2021) opening angles, with the parameters of these fits reported in Table 8. Also reported in Table 8 are the median Classes 0 and I opening angles in the Habel et al. (2021) sample, after applying correction methods 1 and 2. These results show that the two samples are consistent on both the widening between Classes 0 and I outflows and the lack of a discernible continuation of the widening trend between the Class 0/I boundary, regardless of the specific correction method used, emphasizing that the MASSSES and Habel et al. (2021) results for outflow opening angle trends are in overall good agreement. We note that correction method 2 yields opening angles that are overall more consistent with the MASSSES opening angles, especially in the relative lack of very narrow Class I outflows. Finally, comparing the two correction methods emphasizes that the large number of very narrow Class I outflows reported by Habel et al. (2021) is a direct result of their large, model-derived, outflow width reductions by factors of 2–2.3 in outflows with full opening angles less than 30°.

Finally, we note that Habel et al. (2021) used the fraction of point sources (assumed to be viewed face-on through the cleared outflow cavities) to suggest a maximum intrinsic outflow opening angle of 70° in Class I sources, in good quantitative agreement with our results for the existence and possible values of a maximum outflow opening angle ($\sim 66^\circ$ after inclination correction; see Section 5.3.3).

5.1.6 Comparison to Hsieh et al. (2023)

Hsieh et al. (2023) measured outflow opening angles, using ~ 1 arcsec resolution ALMA ^{12}CO (2–1) data, for 17 protostellar outflows in Orion, using a method identical to our own except implemented separately for the redshifted and blueshifted outflow lobes. Since we fit to the wider of the two lobes in the small number of cases where the two outflow lobes are asymmetrical, for consistency with our results we select the wider of their two opening angle measurements.⁵ We also select their results uncorrected for inclination for direct comparison with our results, which are also uncorrected for inclination.

The bottom right panel of Fig. 8 plots the outflow opening angles versus the T_{bol} of the driving sources for both our MASSSES measurements and those presented by Hsieh et al. (2023). Overplotted are unweighted hyperbolic tangent fits to just the MASSSES opening angles, to just the Hsieh et al. (2023) opening angles, and to the combination of the MASSSES and Hsieh et al. (2023) opening angles. The parameters of these fits are reported in Table 8. Also reported in Table 8 are the median Classes 0 and I opening angles in the Hsieh et al. (2023) sample.

Based on their measurements, Hsieh et al. (2023) concluded there is a trend of increasing opening angle with T_{bol} , with a notable decline in the rate of increase for more evolved sources. Overall their results are highly consistent with our MASSSES results.

⁴While Habel et al. (2021) do not tabulate their specific correction factors, based on the information they report in their text we apply correction factors of 1.3 (2.0) for unipolar cavities with reported half-angles greater than (less than) 15°, and correction factors of 1.1 (2.3) for bipolar cavities with reported half-angles greater than (less than) 15°.

⁵The mean and median differences between the wider and narrower lobes for the Hsieh et al. (2023) sample are 19.5° and 20°, somewhat beyond the high end of the 5°–15° typical uncertainty for our method that we discussed in Section 3.2. When using the narrower lobes rather than the wider lobes, the median opening angles reported in table 8 for the Hsieh et al. (2023) outflows decrease from 65° to 42° for Class 0, and from 110° to 85° for Class I.

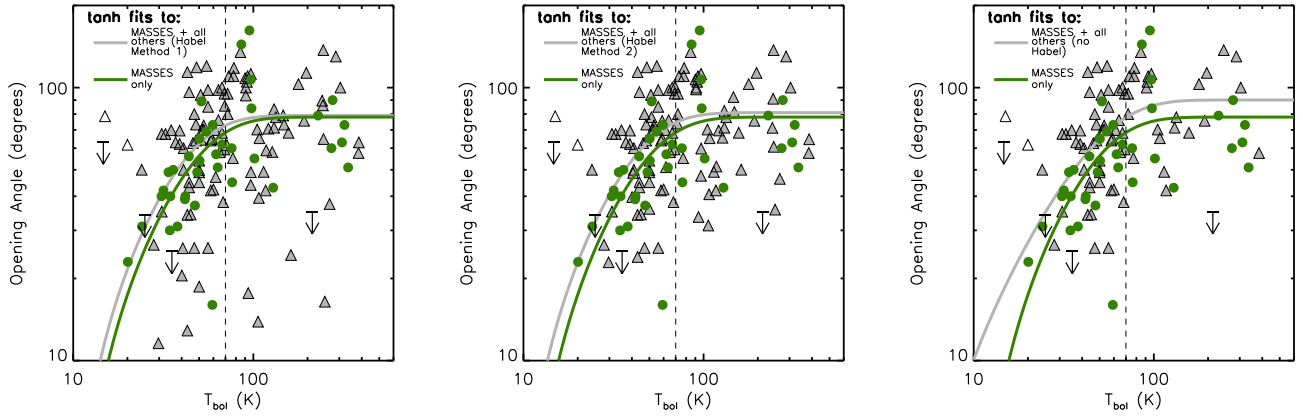


Figure 9. Outflow opening angles plotted versus the bolometric temperatures of their driving sources, on logarithmic axes. In all three panels, the green circles plot the MASSES HC and MC measurements and the downward arrows plot the MASSES upper limits. The shaded grey triangles plot the combined measurements from the previous studies discussed in Section 5.1, with the Habel et al. (2021) measurements corrected using method 1 in the left panel, corrected using method 2 in the middle panel, and omitted from the comparison in the right panel (see the text in Section 5.1.7 for details on the unshaded grey triangles). The dashed vertical line in each panel marks $T_{\text{bol}} = 70$ K, the Class 0/I boundary in bolometric temperature (Chen et al. 1995), with Class 0 protostars to the left ($T_{\text{bol}} < 70$ K) and Class I protostars to the right ($T_{\text{bol}} \geq 70$ K). The green line in each panel shows the same unweighted hyperbolic tangent fit to the MASSES measurements that is shown in panel (d) of Fig. 7, and the grey line in each panel shows the unweighted hyperbolic tangent fit to the combined sample of MASSES measurements and previous literature measurements. Upper limits are not used in any of the fits presented here.

5.1.7 Synthesis of comparisons to previous results

Fig. 9 plots the projected outflow opening angles versus the T_{bol} of the driving sources for the combined MASSES (this work), Velusamy et al. (2014), Hsieh et al. (2017, 2023), Dutta et al. (2020), and Habel et al. (2021) samples. In each panel, the MASSES measurements are plotted with green circles and the combined sample of previous measurements are plotted with filled grey triangles. The Habel et al. (2021) measurements are corrected using method 1 in the left panel, they are corrected using method 2 in the centre panel, and they are omitted entirely in the right panel.⁶ Inspection of Figs 8 and 9 reveals two outliers at low T_{bol} and wide opening angle, one from Hsieh et al. (2017) and one from Dutta et al. (2020). These objects are plotted as unshaded triangles in Fig. 9. The outlier in Hsieh et al. (2017), DCE 004, is undetected in their near-infrared images and only weakly detected in their mid-infrared image, with their best-fitting model image not clearly providing a good fit to their observed image. For the outlier in Dutta et al. (2020), G203.21–11.20W2, Dutta et al. report a T_{bol} of 15 K but this appears to be a mistake; recalculating T_{bol} using their tabulated photometry and stated signal-to-noise cut-offs yields a T_{bol} of 109 K. Thus, we omit these two outliers from the following analysis.

Overplotted in each panel of Fig. 9 are unweighted hyperbolic tangent fits to just the MASSES opening angles and to the combined sample of MASSES and all previous results (except for those omitted as described above), with the parameters of these fits reported in Table 8. Also reported in Table 8 are the median Classes 0 and I opening angles for the combined samples. Regardless of the specific method used to correct the Habel et al. (2021) measurements (or even if the Habel et al. measurements are included at all), the combined sample gives generally consistent results to those obtained with just the MASSES sample: there is a moderately increasing opening angle trend in Class 0 but no clear evidence that this trend continues in Class I, with Class 0 correlation coefficients (in log–log space)

of 0.39 (left panel), 0.41 (middle panel), and 0.39 (right panel), and Class I correlation coefficients (in log–log space) of 0.06 (left panel), 0.06 (middle panel), and 0.07 (right panel). Additionally, the maximum Class I opening angles reached by the hyperbolic tangent fits are 79° (left panel), 81° (middle panel), and 90° (right panel), in good agreement with the value of 78° obtained from the MASSES measurements alone. The scatter for the combined sample is larger, as expected given the use of different outflow tracers and different measurement techniques, but there is overall consistency in the results between the various studies to date. Larger samples using a consistent tracer and uniform measurement technique are needed to further quantify the exact nature of opening angle trends, and indeed such samples should become available in the coming years through the ALMA data archive.

5.2 Implications for the Growth of Protostars

Synthesizing our results and those of the previous studies described above in Section 5.1, Class 0 outflow opening angles show a widening trend with T_{bol} . If T_{bol} is taken as a proxy for evolutionary status, outflows do appear to widen considerably throughout the Class 0 evolutionary stage. Class I outflows, on the other hand, do not show any significant trends between opening angles and T_{bol} . The transition from significant widening to no significant widening appears to occur very near the Class 0/I boundary, with approximately 90 per cent of the widening occurring by the Class 0/I boundary of $T_{\text{bol}} = 70$ K [based on both hyperbolic tangent and $(1 - e)$ fits to the data].

The maximum projected opening angle reached in the Class I stage before the widening ends remains somewhat uncertain. Hyperbolic tangent and $(1 - e)$ fits to just our MASSES data asymptote to opening angles ranging between 65° – 78° , depending on the exact details of the fit. Unweighted hyperbolic tangent fits to various combinations of opening angles from MASSES and other previous studies asymptote to opening angles ranging between 60° – 108° (with mean and median values of 81° and 79° , respectively), and a fit to the combined measurements from MASSES and all previous surveys asymptote to opening angle of 79° – 90° , depending on exactly which measurements are included and exactly how they are corrected for

⁶Including a panel where the Habel et al. (2021) measurements are omitted should not be taken to signify that we distrust their results; rather it is used to further demonstrate the robust nature of the results of this paper.

different measurement techniques. Considering all of these results, we conclude that the maximum projected opening angle reached during the Class I stage is approximately $90^\circ \pm 20^\circ$, where the uncertainty of 20° represents the approximate range of possible values rather than a statistical uncertainty.

Bipolar conical outflow cavities with opening angles ranging from 70° to 110° subtend between 18 per cent–43 per cent of the full 4π steradian. Assuming spherical symmetry, and further assuming that only mass which starts within the outflow cavity can be removed by the outflow, completely cleared conical cavities of these sizes could remove no more than 18 per cent–43 per cent of the total initial core mass. This range represents an upper limit for two reasons: (1) the cavities start much narrower and only reach these widths near the Class 0/I transition, allowing some mass that is within the final outflow cavity to accrete before the cavities reach their full extents; and (2) rotational flattening decreases the fraction of total core mass near the rotation axes (and thus within the volume occupied by the outflow) and increases it near the mid-plane, compared to the fractions expected in spherically symmetric cores. Thus, if it is assumed that the fraction of mass removed by outflows is equal to the fraction of the total volume occupied by conical outflows with these maximum projected opening angles, then these outflows are incapable of removing more than, at most, a few tens of per cent of the initial core mass (and even this is likely a strong upper limit). In this regard we are in qualitative agreement with Habel et al. (2021), who concluded that the low star formation efficiency inferred for molecular cores cannot be explained solely by outflow-induced envelope clearing. On the other hand, Myers et al. (2023) recently showed that outflows that reach maximum projected opening angles of 90° can remove larger mass fractions of the initial core than those calculated here by considering the effects of conical versus curved outflow cavity shapes and/or the possibility that mass which starts outside of the outflow can still be ejected by the outflow (in particular by MHD winds; e.g. Machida & Hosokawa 2013). Thus at present the extent to which outflows are capable of setting the observed star formation efficiencies of 25 per cent–50 per cent on dense core scales, requiring that the outflows remove 50 per cent–75 per cent of the initial core mass, remains uncertain.

5.3 Limitations of the current work

5.3.1 Opacity and chemistry effects

As discussed extensively in Section 5.1.2, opacity and/or chemistry effects may cause outflow opening angle measurements made using ^{12}CO emission to underestimate the intrinsic opening angles. While such an underestimate is unlikely to change the conclusion that a maximum outflow opening angle appears to exist, it may change the quantitative result for the value of this maximum angle. At present, the full magnitude and extent of this issue is unknown due to there being too few outflows sufficiently characterized to perform a statistical comparison of the results obtained using different tracers. Future work must be devoted to measuring the opening angles of the same outflows using ^{12}CO , ^{13}CO , and infrared scattered light images, using consistent techniques applied to each data set, in order to investigate the full magnitude and extent of this issue.

5.3.2 Possible bias against the widest outflows

All investigations of outflow opening angles, whether they use molecular gas observations or scattered light images, must start with

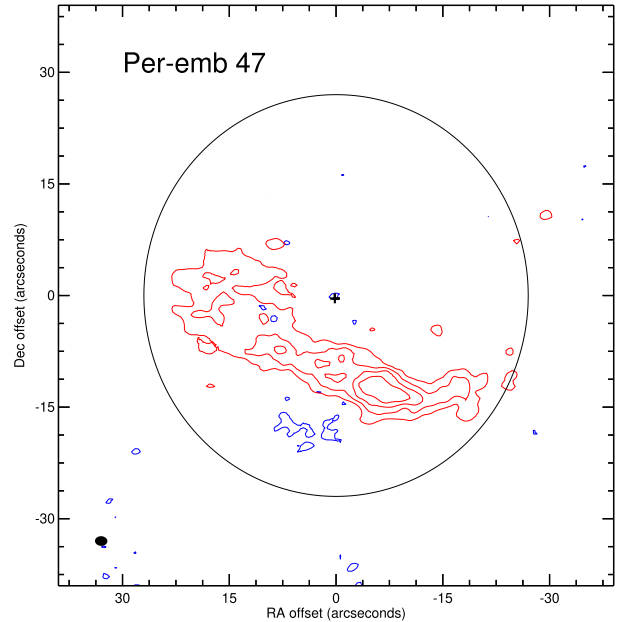


Figure 10. Integrated (moment 0) redshifted and blueshifted ^{12}CO ($2-1$) emission associated with Per-emb 47. The emission is integrated from 2 to 10 km s^{-1} relative to the systemic velocity of 7.5 km s^{-1} . The contours start at 3σ and increase by 3σ , where σ is the rms noise in the integrated maps. The large circle shows the primary beam of the SMA at 230 GHz, centred on the phase centre of the observations. The synthesized beam is shown as the filled ellipse in the lower left corner.

a visual identification of the outflows to be measured. It is possible that all such results, including those presented here, suffer from a systematic bias against the widest outflows. Outflows with opening angles that approach 180° may be difficult to identify since they may not look much like the canonical picture of an outflow, and additionally wide outflows may suffer from incomplete recovery in our interferometric MASSES data due to limited uv -coverage, especially at the shortest uv spacings.

Although cases like Per-emb 36 do suggest confidence in the claim that we are able to identify outflows with very wide opening angles near their bases when they are present (see in particular the blueshifted emission associated with Per-emb 36 in Fig. 5), we do find some cases in our MASSES observations where there is high-velocity gas associated with Class I protostars that we are unable to conclusively identify as an outflow. Fig. 10 shows one such example for Per-emb 47. As noted in Appendix C, we detect substantial redshifted emission in the vicinity of Per-emb 47 in our MASSES ^{12}CO ($2-1$) observations, extending up to $\sim 10 \text{ km s}^{-1}$ from rest. The morphology of this emission is not clearly indicative of an outflow. However, it is difficult to account for gas moving at up to 10 km s^{-1} from rest without invoking outflows, and comparison to larger-scale single-dish maps does support the possibility that this emission is attributed to an outflow that is poorly recovered in our MASSES data (Hatchell & Dunham 2009, see in particular their fig. 5). Future high-sensitivity, high-fidelity observations, especially in the Class I stage, must carefully revisit this question.

5.3.3 Effects of inclination

For outflows viewed at inclination angles of $i = 90^\circ$, corresponding to an edge-on geometry where the outflow axis is aligned with the

plane of the sky, the measured opening angle will be identical to the intrinsic opening angle. As the inclination angle decreases toward 0° (toward a more pole-on orientation), the measured opening angle will be larger than the intrinsic opening angle. Our results are not corrected for inclination since, in many cases, the inclinations of the systems are either unknown or highly uncertain. As a result, most of our measured, projected opening angles will be wider than the intrinsic, deprojected opening angles of the outflows. While it is important to note that neither the lack of wide Class 0 outflows nor the existence of a maximum Class I outflow opening angle can be explained by inclination effects, since inclination corrections would shift our measured opening angles down rather than up, our inability to correct for inclination will likely at least somewhat affect our quantitative results.

We utilize the following Monte Carlo simulation to better quantify the effects of excluding inclination corrections on our results. For each of the 37 HC and MC outflows, we randomly select an inclination between $i = 0^\circ$ (pole-on) and $i = 90^\circ$ (edge-on), weighted by solid angle so that higher inclination angles (more edge-on geometries) are more likely to be selected. We then calculate the inclination-corrected opening angle for each of the 37 outflows (using equation D4), and then refit the unweighted hyperbolic tangent fit described by equation (3). We then repeat this for a total of 10 000 iterations.

The distribution of the parameter θ_0 , which gives the maximum value to which the hyperbolic tangent fit asymptotes (and thus the maximum projected opening angle reached by Class I outflows), is shown in the top panel of Fig. 11. θ_0 has a mean value of 66.0° and a standard deviation of 4.1° over the 10 000 simulation runs. Our original value for θ_0 from Table 6 was 78° , and the smaller mean value here makes sense given that inclination corrections will decrease the measured opening angles. It also agrees very well with the value of 68.7° for θ_0 that is obtained by correcting $\theta_0 = 78^\circ$ after adopting the average inclination angle of 57.3° for randomly oriented objects. The distribution of T_{bol} values where the hyperbolic tangent fit reaches 90 per cent of its maximum value (90 per cent of θ_0) from the 10 000 iterations is shown in the bottom panel of Fig. 11. The mean T_{bol} at which 90 per cent of the widening has occurred is 76.6 K, with a standard deviation of 7.3 K (compared to 74 K for our original fit without inclination corrections).

A Monte Carlo simulation such as this could in theory miss the effects of the most extreme inclination corrections, although with 10 000 simulation runs the statistics are likely robust. We can explicitly test this by considering Per-emb 44 (SVS 13A), which has a robust inclination measurement of $i = 18 \pm 3^\circ$ (Hodapp & Chini 2014) and is thus very close to a pole-on inclination. The large inclination correction necessary for $i = 18^\circ$ reduces the opening angle of this outflow from 144° (the second largest value in our sample) to 87° . Repperforming our unweighted hyperbolic tangent fit to our full sample with the Per-emb 44 opening angle changed to 87° reduces θ_0 (the maximum value to which the hyperbolic tangent fit asymptotes) from 78° to 76° .

Given these results, we find that the inclusion of inclination corrections does not affect our two main conclusions that there exists a maximum outflow opening angle reached in the Class I stage, and that the transition from widening to constant outflow opening angles occurs at or near the Class 0/I boundary. It does reduce the value of the maximum intrinsic opening angle reached in the Class I stage, compared to the projected value. We note that this value is close to the low end of our stated range of approximately $90^\circ \pm 20^\circ$ for the maximum projected opening angle.

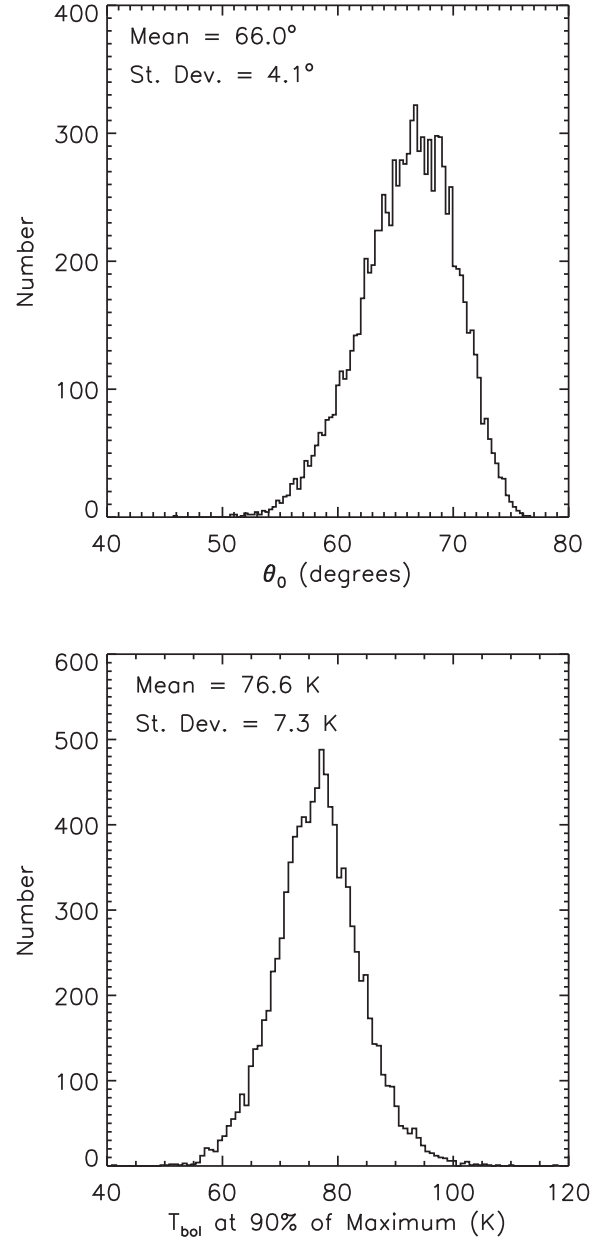


Figure 11. Results from our Monte Carlo test of the effects of inclination corrections on our results. The top panel shows the distribution of the parameter θ_0 from the 10 000 unweighted hyperbolic tangent fits. The bottom panel shows the distribution of T_{bol} values where each of the 10 000 hyperbolic tangent fits reach 90 per cent of their maximum (90 per cent of θ_0).

5.3.4 Use of T_{bol} as an evolutionary indicator

As noted in Introduction, the T_{bol} of a protostar is defined to be the temperature of a blackbody with the same flux-weighted mean frequency as the SED of that protostar (Myers & Ladd 1993), and T_{bol} is commonly used as an evolutionary indicator (e.g. Chen et al. 1995; Enoch et al. 2009; Evans et al. 2009; Dunham et al. 2014a; Frimann et al. 2016). However, in addition to evolutionary status, T_{bol} can also be affected by geometry and accretion history, thus assuming a monotonic increase in T_{bol} with evolutionary status (as is done when using plots of opening angle versus T_{bol} as proxies for plots of opening angle versus evolutionary status) adds potential

uncertainties and biases to the results presented here that are difficult to fully quantify.

Several authors have argued that alternative classification methods, such as the ratio of submillimetre (longward of $350\ \mu\text{m}$) to bolometric luminosity ($L_{\text{submm}}/L_{\text{bol}}$), better trace the underlying evolutionary status of a protostar (e.g. Dunham et al. 2014a; Frimann et al. 2016). Our choice to use T_{bol} is motivated primarily by a desire to remain consistent with previous studies (for ease of comparison) that also used T_{bol} . Future work must examine the extent to which the results of this paper hold up to alternative evolutionary indicators.

6 SUMMARY

In this paper, we use ^{12}CO ($2-1$) observations from the MASSES project to measure the projected opening angles of protostellar outflows in the Perseus Molecular Cloud. We summarize our results as follows:

(i) We measure the opening angles for a total of 46 outflows, 37 of which are measured with sufficiently HC to use for further analysis. Out of these 37 outflows, 23 are driven by Class 0 protostars and 14 are driven by Class I protostars.

(ii) We measure the outflow opening angles using a fitting method adapted from Offner et al. (2011). This method shows excellent agreement with the cones that would be drawn by hand for conical outflows, and it accurately measures the opening angles of the conical (or approximately conical) bases of outflows with curved morphologies.

(iii) There is a statistically significant difference in the distributions of outflow opening angles for Classes 0 and I outflows. In particular, there is a distinct lack of both wide-angle Class 0 outflows and highly collimated Class I outflows, and the median Class I outflow (73°) is wider than the median Class 0 outflow (49°).

(iv) Class 0 outflows have opening angles that exhibit a moderate positive linear correlation with T_{bol} , whereas Class I outflows have opening angles that exhibit either no, or a very weak negative, linear correlation with T_{bol} . If T_{bol} is taken to be an evolutionary tracer, Class 0 outflows appear to widen with age whereas Class I outflows do not.

(v) Multiple types of fits to the opening angles versus T_{bol} , including power-law fits (linear fits in log–log space), weighted and unweighted hyperbolic tangent fits in linear space, and weighted and unweighted exponential fits in linear space, all demonstrate that a transition between outflow opening angles that widen with T_{bol} (widen with age) and outflow opening angles that remain approximately constant with T_{bol} (approximately constant with age) occurs at or near the Class 0/I boundary. These results are seen to be robust even when inclination corrections are taken into account.

(vi) The results from this survey are generally consistent with those from all previously published studies of outflow opening angles over the last 15 yr.

(vii) Outflows do not widen all the way to 180° , at which point they would fully disperse the surrounding core. Instead they appear to reach maximum projected opening angles of approximately $90^\circ \pm 20^\circ$. The volume fractions occupied by these outflows are incapable of removing more than a few tens of per cent of the initial core mass, at most, assuming the fraction of initial core mass removed by the outflow matches the volume fraction of the core occupied by these conical outflows.

While these results challenge the notion that outflows widen all the way to 180° , at which point they disperse the surrounding cores and terminate the protostellar mass accretion process, recent

theoretical work suggests outflows may still be capable of playing a central role in setting the low star formation efficiencies of 25 per cent–50 per cent observed on core scales. Future work must concentrate on building larger samples with improved statistics, especially in the Class I stage, on understanding the systematic biases introduced by measuring outflow opening angles with different data sets and methodologies, and on testing the effects of using alternative evolutionary indicators to measure the relative evolutionary status of the driving sources.

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DATA AVAILABILITY

All MASSES data are publicly available through the Harvard Dataverse at https://dataverse.harvard.edu/dataverse/full_MASSES/.

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APPENDIX A: OUTFLOWS IN CLUSTERS AND GROUPS

In this appendix, we present we present large-scale maps showing the results of our individual MASSES pointings in various regions of Perseus. Figs A1–A5 show the IC348, B1, NGC 1333, L1455, and L1448 regions of Perseus, respectively. Individual protostars are labelled as PerXX instead of Per-emb XX for maximum clarity in the figures. We emphasize that these are not mosaics. Instead, these figures plot the individual pointings in each region offset from each other by the proper amounts to reflect their positions relative to each other. The choice of whether to display the robust = +1 or robust = −1 map for each pointing matches the rest of the paper, as summarized in Table 1. For pointings with undetected or unclear outflows that are not considered in this work, we default to the robust = +1 map. By plotting in offset coordinates there are technically projection errors in these maps in that the decrease in the total distance across the right ascension axis by a factor of $\cos(\delta)$ with increasing δ (where δ represents the declination) is not accounted for. However, the total change in distance across the right ascension axis between the bottom and top of these figures is less than 1 arcsec, and thus negligible, for all but NGC 1333. This total change is

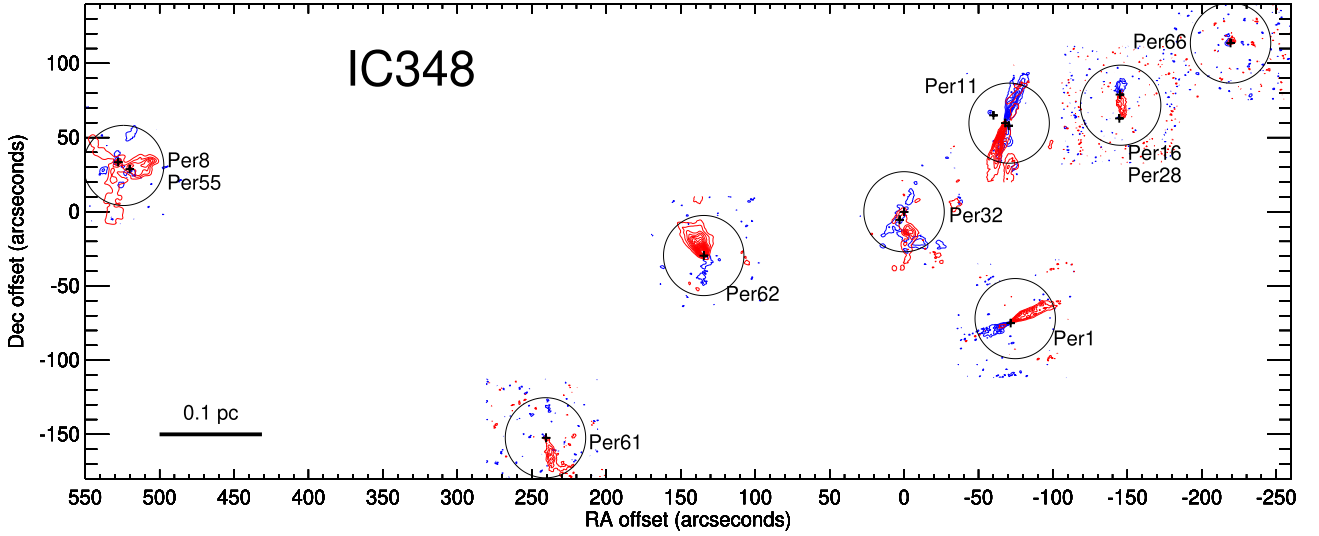


Figure A1. ^{12}CO (2 – 1) outflow maps for each of the MASSES pointings in the IC348 region of Perseus, with the protostars located within each pointing labelled in the figure. The blue contours show blueshifted emission integrated from 3 to 8 km s^{-1} relative to the systemic velocity of 8.5 km s^{-1} , and the red contours show redshifted emission integrated over the same velocity range relative to the systemic velocity. The contours start at 2.5σ and increase by 5σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the pointings. The crosses mark the positions of the protostars. See Figs 3–5 for labels of the individual protostars within each pointing, as well as ellipses showing the size and orientation of each synthesized beam (which would be too small to be visible in this figure). The axes are plotted using coordinate offsets from the phase centre of Per-emb 32 (RA 03:44:02.40 and Dec. +32:02:04.90).

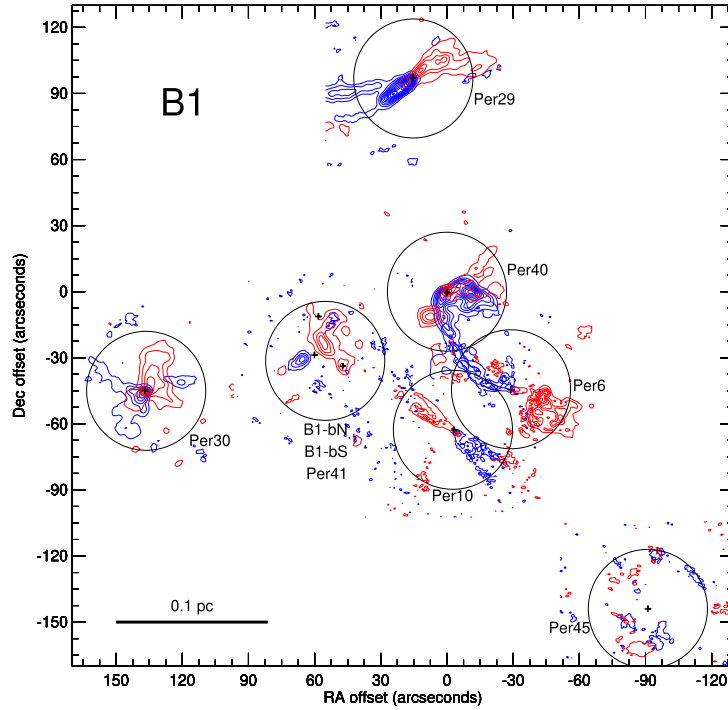


Figure A2. ^{12}CO (2 – 1) outflow maps for each of the MASSES pointings in the B1 region of Perseus, with the protostars located within each pointing labelled in the figure. The blue contours show blueshifted emission integrated from 3 to 8 km s^{-1} relative to the systemic velocity of 6.5 km s^{-1} , and the red contours show redshifted emission integrated over the same velocity range relative to the systemic velocity. The contours start at 2σ and increase by 5σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the pointings. The crosses mark the positions of the protostars. See Figs 3–5 for labels of the individual protostars within each pointing, as well as ellipses showing the size and orientation of each synthesized beam (which would be too small to be visible in this figure). The axes are plotted using coordinate offsets from the phase centre of Per-emb 40 (RA 03:33:16.66 and Dec. +31:07:55.20).

NGC 1333

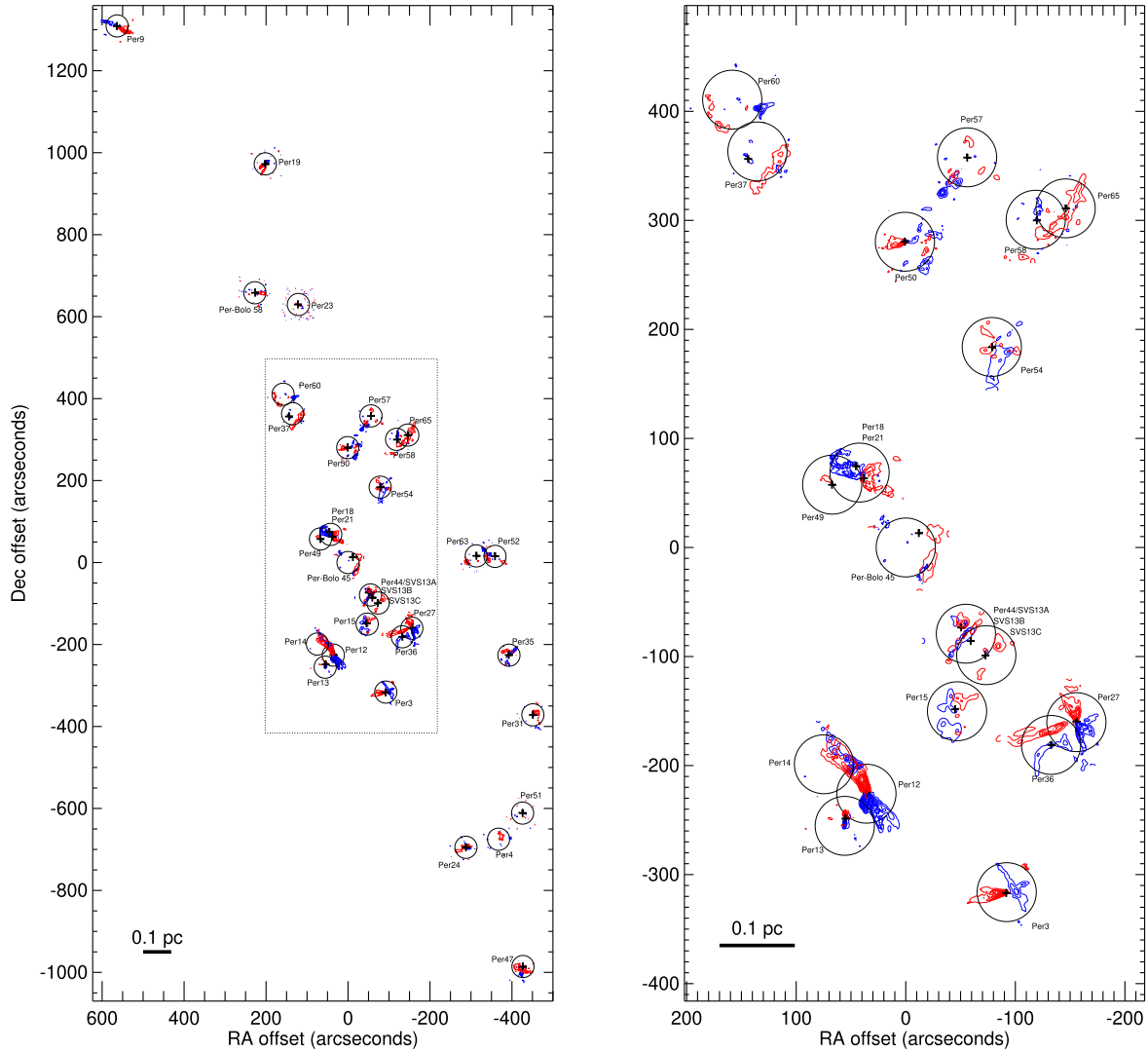


Figure A3. ^{12}CO (2 – 1) outflow maps for each of the MASSES pointings in the NGC 1333 region of Perseus, with the protostars located within each pointing labelled in the figure. The blue contours show blueshifted emission integrated from 3 to 8 km s^{-1} relative to the systemic velocity of 7.5 km s^{-1} , and the red contours show redshifted emission integrated over the same velocity range relative to the systemic velocity. The contours start at 3σ and increase by 5σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the pointings. The crosses mark the positions of the protostars. See Figs 3–5 for labels of the individual protostars within each pointing, as well as ellipses showing the size and orientation of each synthesized beam (which would be too small to be visible in this figure). The axes are plotted using coordinate offsets from the phase centre of Per-Bolo 45 (RA 03:29:07.70 and Dec. +31:17:16.80). The right panel zooms-in on the region outlined with a dotted rectangle in the left panel.

14 arcsec for NGC 1333; while still small compared to the 57 arcsec FWHM (full width at half-maximum) of the primary beam of our observations, it should nevertheless be considered when comparing to maps plotted in absolute coordinates.

In these figures, the velocity ranges for integration and the contour levels plotted do not match those adopted for Figs 3–5, where they

were both customized on a source-by-source basis to best display the outflows in each pointing. Here, we adopt the same velocity range and contour levels for each pointing in a given region, since these figures are primarily intended to give context on the broader region in which each pointing is located, and we refer to Figs 3–5 for maps optimized to show each detected outflow.

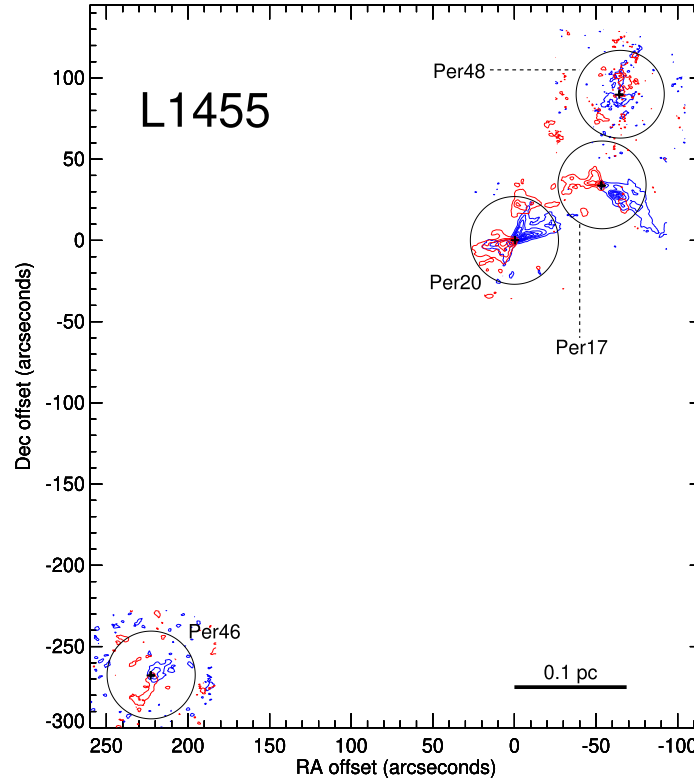


Figure A4. ^{12}CO (2 – 1) outflow maps for each of the MASSES pointings in the L1455 region of Perseus, with the protostars located within each pointing labelled in the figure. The blue contours show blueshifted emission integrated from 3 to 8 km s^{-1} relative to the systemic velocity of 5.5 km s^{-1} , and the red contours show redshifted emission integrated over the same velocity range relative to the systemic velocity. The contours start at 2σ and increase by 5σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the pointings. The crosses mark the positions of the protostars. See Figs 3–5 for labels of the individual protostars within each pointing, as well as ellipses showing the size and orientation of each synthesized beam (which would be too small to be visible in this figure). The axes are plotted using coordinate offsets from the phase centre of Per-emb 20 (RA 03:27:43.23 and Dec. +30:12:28.80).

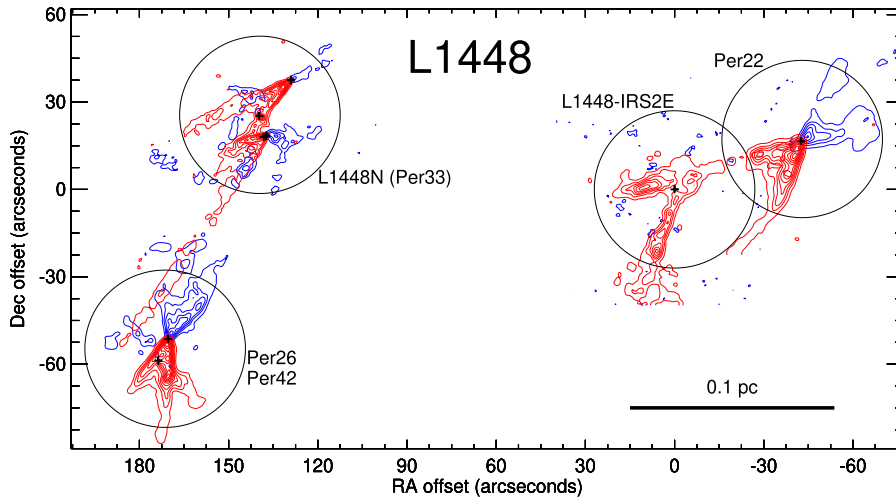


Figure A5. ^{12}CO (2 – 1) outflow maps for each of the MASSES pointings in the L1448 region of Perseus, with the protostars located within each pointing labelled in the figure. The blue contours show blueshifted emission integrated from 2 to 6 km s^{-1} relative to the systemic velocity of 5 km s^{-1} , and the red contours show redshifted emission integrated over the same velocity range relative to the systemic velocity. The contours start at 2.5σ and increase by 5σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the pointings. The crosses mark the positions of the protostars. See Figs 3–5 for labels of the individual protostars within each pointing, as well as ellipses showing the size and orientation of each synthesized beam (which would be too small to be visible in this figure). The axes are plotted using coordinate offsets from the phase centre of L1448-IRS2E (RA 03:25:25.66 and Dec. +30:44:56.70).

APPENDIX B: NOTES ON INDIVIDUAL SOURCES WITH MEASURED OPENING ANGLES

In this appendix, we provide relevant notes on each outflow for which we obtain an opening angle measurement. We note here that we applied our fitting method to both outflow lobes simultaneously in all cases except for those specifically noted below.

Per-emb 1. The Per-emb 1 outflow is well detected in our MASSES ^{12}CO (2 – 1) observations. While it is not well resolved in our robust +1 image, it is well resolved in our robust –1 image. This outflow is well fitted by our method, thus we place it in the HC group.

Per-emb 2. Per-emb 2 is well fitted by our method, thus we place this object in the HC group. While Per-emb 2 is a binary system with a projected separation of 24 au, as revealed by the VANDAM (Tobin et al. 2016, 2018), we see no evidence of multiple outflows and thus assume that one component dominates both the outflow and the observed SED.

Per-emb 3. The Per-emb 3 outflow is well detected in our MASSES ^{12}CO (2 – 1) observations. While some of the low-velocity blueshifted channels (notably those at $\sim 3 \text{ km s}^{-1}$ from rest) are affected by striping due to poor *uv* coverage, we are able to obtain a reliable fit to by integrating over higher velocities. Given the clear outflow signature and the extent to which our fit agrees visually with the morphology of this outflow, we place this object in the HC group.

Per-emb 5. Per-emb 5 is well fitted by our method, thus we place this object in the HC group. As with Per-emb 2, Per-emb 5 is a binary system with a projected separation of 29 au (Tobin et al. 2016, 2018). With no evidence of multiple outflows in our data, we again assume that one component dominates both the outflow and observed SED.

Per-emb 6. The Per-emb 6 outflow is well detected in our MASSES ^{12}CO (2 – 1) observations. Despite being located in the B1 region where the CO emission is quite complex, this outflow is sufficiently isolated for identification and fitting (see Fig. A2). While our fitting method provides a good visual match to the redshifted lobe, the match to the blueshifted lobe is somewhat more questionable, possibly due to contamination by other outflows in the B1 complex. We thus place this object in the MC group.

Per-emb 7: Per-emb 7 is associated with a highly collimated, redshifted outflow lobe that extends to the northwest of the protostar, but the blueshifted emission is significantly more complex. While there are some hints of a similarly collimated blueshifted lobe extending to the southeast, the emission is dominated by a diagonal, southwest–northeast stripe that is not clearly associated with Per-emb 7 itself. As seen in Fig. B1, this stripe is aligned with the blueshifted lobe of the outflow driven by Per-emb 34, located $4.5'$ to the southwest. Thus, it is likely that the blueshifted emission in the vicinity of Per-emb 7 is contaminated by the outflow from Per-emb 34, as also noted by Hatchell & Dunham (2009). As a result, we fit to only the redshifted outflow lobe. Given that this lobe is only marginally resolved in our data, our measured opening angle is an upper limit to the actual opening angle (we are unable to use the higher resolution robust –1 image for this outflow due to insufficient sensitivity).

Per-emb 9. The Per-emb 9 outflow is well detected in our MASSES ^{12}CO (2 – 1) observations, where it displays an S-shaped symmetry in both lobes suggesting possible outflow precession. While Per-emb 9 is single down to the VANDAM resolution limit of 20 au (Tobin et al. 2016, 2018), it is possible there is an undetected close companion responsible for this apparent precession. Additionally, the redshifted lobe features a northwestern extension of unknown origin approximately halfway down its length, but since this extension does

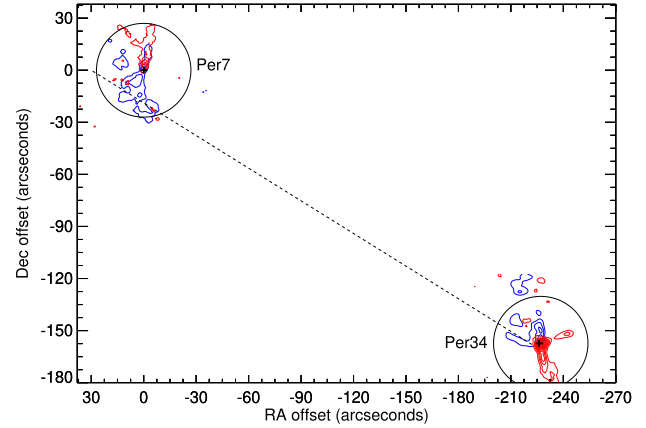


Figure B1. Integrated redshifted and blueshifted ^{12}CO (2 – 1) emission associated with Per-emb 7 and Per-emb 34, integrated over the velocity ranges listed in Table 1. The contours start at 2.5σ and increase by 3σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the two pointings. The dashed line is drawn to guide the eye, and indicates that the blueshifted outflow lobe from Per-emb 34 is directed toward Per-emb 7. The synthesized beams for each pointing are shown in Figs 3 and 5 (they are so small they would be nearly invisible on the scale of this figure). The axes are plotted using coordinate offsets from the phase centre of Per-emb 7.

not clearly point back to Per-emb 9, and is only marginally spatially connected to the southwestern redshifted outflow lobe, it is unlikely but ultimately unclear if it is indeed part of the Per-emb 9 outflow. In fitting the opening and position angles, we only fit close to the source (within $15''$) where the curvature in each lobe has not yet become apparent, and we use the higher resolution robust –1 image as the collimated outflow lobes are better resolved in this image. While our fitting method provides a good visual match to the observed outflow morphology, given the uncertainties associated with this outflow, including the possible precession and the redshifted extension of unknown origin, we place Per-emb 9 in the MC group.

Per-emb 10. The Per-emb 10 outflow is well detected in our MASSES ^{12}CO (2 – 1) observations. Like Per-emb 6, despite being located in the B1 region where the CO emission is quite complex, this outflow is sufficiently isolated for identification and fitting (see Fig. A2). A companion object detected to the southeast by both Pokhrel et al. (2018) and Stephens et al. (2018), and shown to be driving an outflow in the latter study, is in fact an artefact due to phase errors in the original data set that have now been corrected (Stephens et al. 2019). While the outflow is not well resolved in our robust +1 image it is well resolved and well fitted in our robust –1 image, thus we place this object in the HC group.

Per-emb 11. Per-emb 11 is a binary system with a projected angular separation of 2.95 arcsec , corresponding to a projected linear separation of 885 au (Tobin et al. 2016, 2018). Both components in this binary system drive nearly aligned, overlapping, bipolar outflows, except with reversed redshifted and blueshifted lobes (the blueshifted lobe extends to the northwest [southeast] for the primary [secondary] source, and the redshifted lobe extends to the southeast [northwest] for the primary [secondary] source). It is worth pointing out that Pech et al. (2012) interpreted these parallel outflows as evidence of jet rotation in lower resolution observations, although they did explicitly acknowledge the possibility of parallel outflows driven by unresolved (at the time) binary components.

Despite the spatial overlap the two outflows are well separated in velocity, allowing us to apply our fitting procedure to the primary

outflow (assumed to be driven by the same source that dominated the observed SED and thus dominates the measured T_{bol}). Since our measured opening angle accurately reproduces the extent of the primary outflow, we place this source in the HC bin. We do not fit to the secondary outflow since we do not have a reliable measurement of T_{bol} for this outflow's driving source.

A third protostar, IC348-SMM2E, is located at a projected separation of 9.47 arcsec to the northeast of the primary component of the Per-emb 11 binary system, corresponding to a projected linear separation of 2841 au (Palau et al. 2014; Tobin et al. 2016, 2018). It drives a compact outflow first detected by Palau et al. (2014), and indeed the blueshifted lobe is visible in Fig. 3. However, as this outflow is unresolved in our data and contained entirely within 4 arcsec of its driving source, we are unable to reliably apply our fitting procedure.

Per-emb 13. A compact outflow associated with Per-emb 13 is well detected in our MASSSES ^{12}CO (2 – 1) observations. The northern redshifted lobe is contaminated by sidelobe emission associated with the nearby Per-emb 12 outflow, thus we fit to only the blueshifted emission. The width of this outflow is unresolved in both the robust +1 and robust –1 images, thus our measured opening angle is an upper limit to the true opening angle.

An additional Class 0 protostar, referred to by various names in the literature including IRAS4C (Looney, Mundy & Welch 2000), IRAS4B2 (Hull et al. 2014), and IRAS4B' (Tobin et al. 2016), is located 10.66 arcsec east of Per-emb 13, corresponding to a projected linear separation of 3196 au (Looney et al. 2000; Tobin et al. 2016, 2018). It drives a weak, compact outflow whose redshifted lobe is weakly detected in our data, but as this outflow is unresolved in both dimensions and contaminated by sidelobe emission from the Per-emb 12 outflow, we are unable to measure its opening and position angles.

Per-emb 15. A bipolar displaying outflow a wide, shell-like morphology, consistent with the ^{12}CO (1–0) map presented by Plunkett et al. (2013), is clearly detected from Per-emb 15. While the outflow morphology is clearly not conical, the opening angle near the source is very wide and our fitting method accurately reproduces this wide angle, leading us to place this object in the HC group.

Per-emb 16. Per-emb 16 is part of the Per-emb 16/28 wide binary system, with a projected separation of 16 arcsec (corresponding to a projected linear separation of 4819 au; Tobin et al. 2016, 2018). The two components drive approximately perpendicular outflows (e.g. Lee et al. 2016), allowing us to easily separate their ^{12}CO (2 – 1) emission. We use the robust –1 image to best separate the two outflows, and we restrict our fitting method to only those pixels within 13 arcsec of the driving source to avoid contamination from the Per-emb 28 outflow. Our fitting method consistently gives a rather wide opening angle that provides a good fit very close to the driving source, but appears to be a poor fit farther from the source. This fact, coupled with possible asymmetries between the two lobes and contamination from the Per-emb 28 outflow, lead us to place this object into the low confidence group.

Per-emb 17. The outflow driven by Per-emb 17 is clearly detected and well fitted by our method, thus we place this object in the HC group. Per-emb 17 is actually a binary system with a projected angular separation of 0.28 arcsec, corresponding to a projected linear separation of 83 au (Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that the same single component dominates both the outflow and the observed SED.

Per-emb 19. While an outflow driven by Per-emb 19 is clearly detected, the two lobes display somewhat different morphologies. The southeast, redshifted lobe appears to trace two cavity walls with

a cleared interior, with an initially wide angle near the protostar and a more collimated appearance at larger distances along the lobe. The northwest blueshifted lobe does not exhibit a clear cavity structure, and is instead dominated by a bright peak near the protostar. As Per-emb 19 is located to the north of NGC 1333, it is possible that the blueshifted lobe is propagating into a less dense part of the cloud where there is less gas to entrain into the outflow, possibly explaining the lack of emission farther along the blue lobe. While we fit to both lobes simultaneously and our fit accurately traces the wide morphology near the driving source (especially prevalent in the redshifted lobe), the asymmetry in the two lobes leads us to place this source in the MC group.

Per-emb 20. A bipolar outflow driven by Per-emb 20 is clearly detected. There is a redshifted emission peak detected to the north of the blueshifted lobe, visible in Fig. 4 and Fig. A4, that appears unrelated to the outflow and is thus excluded from our fit (it may be related to the Per-emb 17 outflow, but we are unable to conclusively determine its origin). As both lobes of this outflow are well fitted by our method, we place this source in the HC group.

Per-emb 21. Per-emb 21 is part of the Per-emb 18/21 wide binary system, with a projected angular separation of 13.3" (corresponding to a projected linear separation of 3976 au; Tobin et al. 2016, 2018). The MASSSES data clearly detect a relatively wide-angle, bipolar outflow driven by Per-emb 21 (see also Lee et al. 2016), but no outflow is detected from Per-emb 18. The Per-emb 21 outflow displays a substantial asymmetry, with the southwestern, redshifted lobe appearing wider and at a somewhat different position angle than the northeastern, blueshifted lobe. We choose to fit to the blueshifted lobe as our method obtains qualitatively better fits for this lobe. Due to the substantial uncertainties associated with this asymmetrical outflow, along with possible confusion from an undetected Per-emb 21 outflow, we place this source in the low confidence group.

Per-emb 22. The outflow driven by Per-emb 22 is clearly detected and well fitted by our method, thus we place this object in the HC group. Per-emb 22 is actually a binary system with a projected angular separation of 0.75 arcsec, corresponding to a projected linear separation of 225 au (Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that the same single component dominates both the outflow and the observed SED.

Per-emb 23. The outflow driven by Per-emb 23 is clearly detected and well fitted by our method, thus we place this object in the HC group. We use the robust –1 image since this outflow is quite collimated and better resolved in this image. The detection of both redshifted and blueshifted emission in each lobe suggests that this source may be viewed at an inclination close to edge-on.

Per-emb 24. The outflow driven by Per-emb 24 is clearly detected and well fitted by our method, thus we place this object in the HC group.

Per-emb 25. The outflow driven by Per-emb 25 is clearly detected and well fitted by our method, thus we place this object in the HC group.

Per-emb 26. Per-emb 26 is part of the Per-emb 26/42 wide binary system, with a projected separation of 8.1 arcsec (corresponding to a projected linear separation of ~ 2431 au; Tobin et al. 2016, 2018). The two components drive approximately perpendicular outflows that are thus able to be separated spatially. Per-emb 26 drives the dominant outflow in this system, with a northwest blueshifted lobe and a southeast redshifted lobe. As Per-emb 42 is located within the redshifted lobe of the Per-emb 26 outflow, at least in projection, and the outflow driven by Per-emb 42 overlaps spatially with this redshifted lobe, we fit to only the blueshifted lobe of the Per-emb 26 outflow. Redshifted emission northeast of the Per-emb 26 protostar

arises from one of the outflows driven by the Per-emb 33 system, as seen in Fig. A5, and is thus omitted from our fit. Given that the two lobes of the Per-emb 26 outflow are symmetrical and the blueshifted lobe is well fitted by our method, we place this object in the HC group.

Per-emb 27. Per-emb 27 is a binary system with a projected separation of 0.62 arcsec (corresponding to a projected linear separation of 186 au; Tobin et al. 2015, 2016, 2018). The two components drive perpendicular outflows (north–south for the primary and east–west for the secondary; Plunkett et al. 2013; Tobin et al. 2015). Since the SEDs of these components are unresolved at most wavelengths we fit only the primary outflow, under the assumption that the primary source dominates the observed SED (and thus the measured T_{bol}). Inspection of our data, as well as the outflow maps presented by Plunkett et al. (2013) and Tobin et al. (2015), reveals an asymmetry between the two lobes, especially in terms of position angle. However, the blueshifted lobe is contaminated by blueshifted emission from both the secondary outflow and the wide-angle outflow driven by Per-emb 36 (see below, as well as Plunkett et al. 2013), thus we only fit to the northern, redshifted lobe. Given the uncertainties associated with both the asymmetry and the contamination from nearby outflows, we place this object in the MC group.

Per-emb 28. Per-emb 26 is part of the Per-emb 16/28 wide binary system, with a projected separation of 16 arcsec (corresponding to a projected linear separation of 4819 au; Tobin et al. 2016, 2018). The two components drive approximately perpendicular outflows (e.g. Lee et al. 2016), allowing us to easily separate their ^{12}CO (2–1) emission. For Per-emb 28, we remove the central pixels where the two outflows overlap to avoid contamination from the Per-emb 16 outflow. There is a change in position angle between the two lobes of this outflow, as also seen in the map presented by Lee et al. (2016), possibly due to interaction between the two outflows. We thus fit to only the northwestern outflow lobe, and note here that a fit to the southeastern lobe yields a different position angle but a consistent opening angle. Due to the uncertainties from asymmetry and potential confusion, we place Per-emb 28 in the MC group.

Per-emb 29. The Per-emb 29 outflow is well detected in our MASSES ^{12}CO (2–1) observations, where it displays an S-shaped symmetry in both lobes suggesting possible outflow precession. While Per-emb 29 is single down to the VANDAM resolution limit of 20 au (Tobin et al. 2016, 2018), it is possible there is an undetected close companion causing this apparent precession. As our method is incapable of fitting an outflow with a variable position angle, we restrict our fit to only the emission located within 15 arcsec of the protostar in right ascension. Given the uncertainties associated with this choice, we place Per-emb 29 in the MC group.

Per-emb 30. The outflow driven by Per-emb 30 is clearly detected and well fitted by our method, thus we place this object in the HC group.

Per-emb 31. The outflow driven by Per-emb 31 is clearly detected in our data. The northwest lobe is associated with redshifted emission that appears to be tracing resolved outflow cavity walls and spatially coincident blueshifted emission that likely indicates a nearly edge-on inclination for this source. While there is blueshifted emission to the southeast at the expected position angle for this outflow, there is no matching redshifted emission. Since the redshifted emission displays the widest morphology, with resolved cavity walls clearly evident, we only fit to the redshifted emission, and we place this source in the MC group due to the uncertainties associated with the asymmetry. We also note that Per-emb 31 is not detected in our MASSES 1.3 mm continuum observations, but spatially coincident emission is detected

at both 8 mm in the VANDAM survey (Tobin et al. 2016) and in the infrared by Enoch et al. (2009).

Per-emb 33. The Per-emb 33 system (also referred to as the L1448N system or the L1448 IRS 3 system) is a hierarchical system of six protostars distributed in a close triple system, a close binary system, and a single system (Lee et al. 2015; Tobin et al. 2016). We are able to identify, separate, and reliably measure opening angles for two outflows in this system: L1448N-NW and L1448N-B. These outflows are discussed in the appropriate paragraphs below.

Per-emb 34. The outflow driven by Per-emb 30 is clearly detected and generally well fitted by our method, although the morphology differs somewhat between the two lobes, with a more conical appearance in the redshifted lobe and a more parabolic appearance in the blueshifted lobe. While our fit provides a good visual match to the image, to be conservative we place this object in the MC group to account for uncertainties due to the asymmetry.

Per-emb 36. Per-emb 36 drives a very wide outflow with a blueshifted lobe clearly detected in our MASSES data that extends to the southwest and exhibits a shell-like morphology (see also Plunkett et al. 2013). There is redshifted emission to the north, but it is associated with the secondary outflow from Per-emb 27 rather than the outflow driven by Per-emb 36. Farther to the north, outside of the extent of our SMA observations, there is redshifted emission that likely arises from both the Per-emb 27 and Per-emb 36 outflows (see Plunkett et al. 2013). While our method provides a good visual match to the width of the outflow near the driving source, which indeed appears to span nearly the full 180° , the lack of a detected redshifted lobe with which to validate our results leads us to place this object in the MC group. Finally, we note that this object is actually a binary system unresolved in our observations, with a projected linear separation of 93 au (Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that one component dominates both the outflow and the observed SED.

Per-emb 40. The outflow driven by Per-emb 40 is most apparent in our MASSES data as a conical redshifted lobe extending to the northwest. Blueshifted emission from the Per-emb 6 outflow overlaps with this object (see Fig. A2), causing difficulty in identifying a clear blueshifted component to the Per-emb 40 outflow. We thus fit to only the redshifted emission after manually cutting out the additional redshifted emission visible to the southeast of the protostar, as we are unable to determine the origin of this emission. Finally, we also note that this object is actually a binary system unresolved in our observations, with a projected linear separation of 117 au (Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that one component dominates both the outflow and the observed SED. Given the various uncertainties associated with this source, we place it in the MC group.

Per-emb 41. Per-emb 41 is part of the B1-b system, located approximately 14 arcsec (4200 au) southwest of B1-bS, and approximately 25 arcsec (7500 au) southwest of B1-bN (Tobin et al. 2016). Per-emb 41 itself is a single system down to a projected separation of at least 20 au (Tobin et al. 2016, 2018). As previously identified by both Hirano & Liu (2014) and Lee et al. (2016), Per-emb 41 drives a northeast–southwest outflow, with blueshifted emission extending to the southwest and redshifted emission extending to the northeast. Both of these lobes are detected in our MASSES data. While the redshifted lobe eventually merges with confused redshifted emission from the other components of the B1-b system, a clear, compact outflow can be identified in both the redshifted and blueshifted lobes. As both lobes are only marginally resolved in their lengths, and essentially unresolved in their widths, we are only able to measure an upper limit to the opening angle of this outflow.

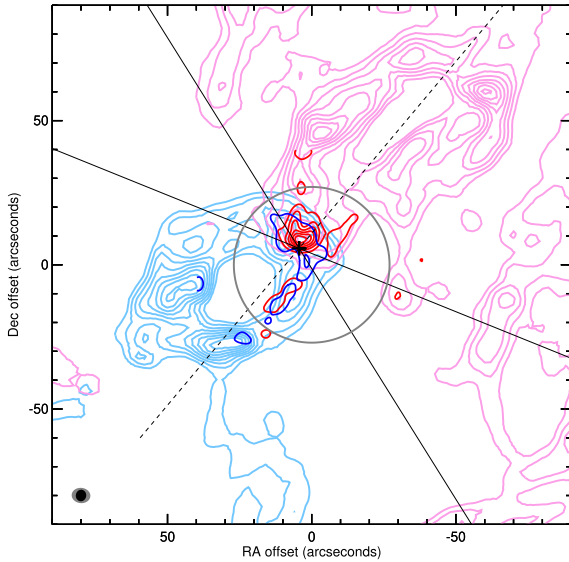


Figure B2. Integrated redshifted (red contours) and blueshifted (dark blue contours) MASSES ^{12}CO (2 – 1) emission associated with Per-emb 44, with the same velocity ranges and contour levels as in Fig. 4. The large grey circle shows the primary beam of the SMA, the cross shows the position of the Per-emb 44 protostar, and the dashed and solid lines show the position and opening angles of the Per-emb 44 outflow from our fit (these are the same lines shown in Fig. 4). Overplotted are redshifted (pink contours) and blueshifted (light blue contours) ^{12}CO (1–0) emission from Plunkett et al. (2013). The synthesized beams for the MASSES (black) and Plunkett et al. (2013) (grey) data are plotted as filled ovals in the lower left corner. The axes are plotted using coordinate offsets from the phase centre of the MASSES observations of Per-emb 44.

Per-emb 42. Per-emb 42 is part of the Per-emb 26/42 wide binary system, with a projected separation of 8.1 arcsec (corresponding to a projected linear separation of ~ 2431 au; Tobin et al. 2016, 2018). The two components drive approximately perpendicular outflows that are thus able to be separated spatially. A northeast–southwest outflow driven by Per-emb 42 is weakly detected in our MASSES data. As the southwestern lobe overlaps substantially with the Per-emb 26 outflow, we only fit to the blueshifted emission in the northeastern lobe (as this lobe overlaps with redshifted emission from outflows driven by the Per-emb 33 system [see Fig. A5], we do not include redshifted emission in our fit). While our fit provides an excellent visual match to the northeastern lobe, the uncertainties associated with the overlapping emission from the multiple outflows lead us to place this object in the MC group.

Per-emb 44. Per-emb 44 drives a wide, bipolar, shell-like outflow, with blueshifted emission extending to the southeast and redshifted emission extending to the northwest (Plunkett et al. 2013, see, in particular, their figs 4 and 5). Only patchy emission from the inner portion of this outflow is detected in our MASSES data. We fit only to the blueshifted lobe since the very wide, shell-like structure is more apparent in this lobe, but we note that a fit to the redshifted lobe gives a similar opening angle. Fig. B2 repeats the Per-emb 44 panel from Fig. 4 except with modified axes ranges to show a larger region and with the ^{12}CO (1 – 0) contours from Plunkett et al. (2013) overplotted. This figure emphasizes that the base of each lobe is well resolved compared to the synthesized beams of the MASSES and Plunkett et al. (2013) data sets, confirming that this outflow is indeed very wide at its base. Due to the uncertainties associated with fitting to the very patchy MASSES ^{12}CO (2 – 1) emission, we place this object in the MC group. Finally, we note that Per-emb 44 is

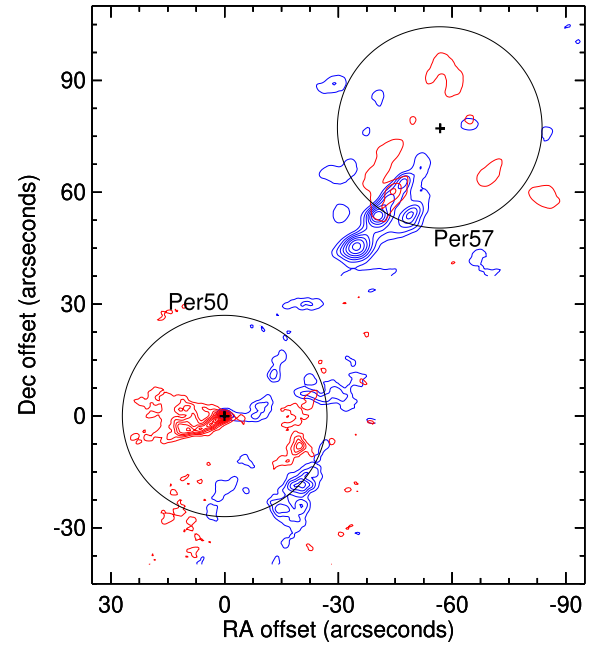


Figure B3. Integrated redshifted and blueshifted ^{12}CO (2 – 1) emission associated with Per-emb 50 and Per-emb 57, integrated over the velocity ranges listed in Table 1. The contours start at 2σ and increase by 3σ , where σ is the rms noise in each integrated map. The large circles show the primary beam of the SMA at 230 GHz, centred on each of the two pointings. The synthesized beams for each pointing are shown in Fig. 5 (they are so small they would be nearly invisible on the scale of this figure). The axes are plotted using coordinate offsets from the phase centre of Per-emb 50.

actually a binary system with a projected separation of 0.3 arcsec (corresponding to a projected linear separation of 90 au Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that one component dominates both the outflow and the observed SED.

Per-emb 46. An outflow driven by Per-emb 46 is clearly detected in our MASSES data, with the blueshifted lobe exhibiting a resolved cavity morphology but only one of the two cavity walls robustly detected in the redshifted lobe. As the ‘missing’ redshifted cavity wall is faintly present in individual velocity channels, and our method provides a good visual fit to the blueshifted lobe, we place this object in the HC group.

Per-emb 50. An outflow driven by Per-emb 50 is clearly detected in our MASSES data. Close to the driving source a classic bipolar outflow oriented east-west is clearly visible. Farther to the south and west there is additional blueshifted emission detected that is not clearly part of this outflow. A larger map reveals that most or all of this additional emission is likely associated with the Per-emb 57 outflow (see Fig. B3), and is thus excluded from our fit. Given the clear outflow signature and the extent to which our fit agrees visually with the morphology of this outflow, we place this object in the HC group.

Per-emb 52. A fairly collimated, bipolar, northeast–southwest outflow driven by Per-emb 52 is detected in our MASSES data, with the northeast lobe robustly detected but the southwest lobe only marginally detected in a few individual velocity channels. While our method does provide a reliable fit to the blueshifted emission, given the overall weak emission and the near total absence of the redshifted lobe, we place this object in the MC group.

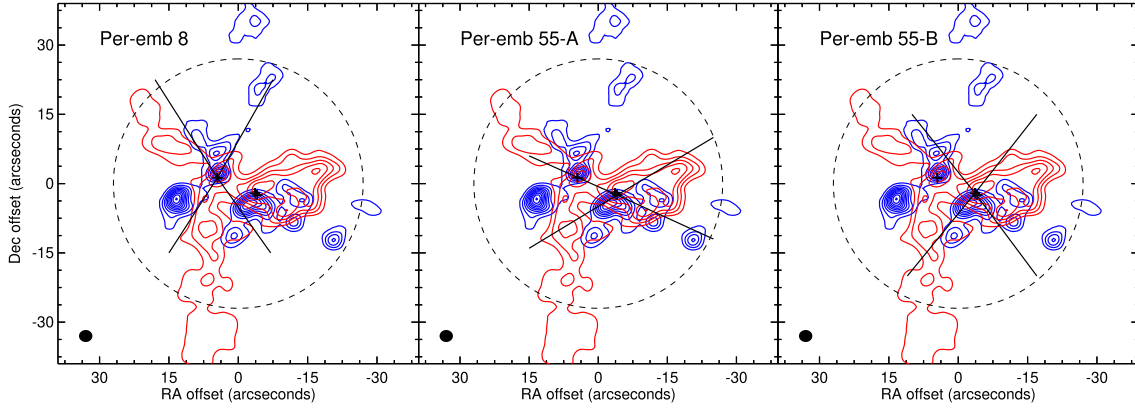


Figure B4. *MASSES* ^{12}CO (2 – 1) integrated emission contours for the Per-emb 8/55 protostellar system. The contours, which are the same in all three panels, show blueshifted (blue contours) and redshifted (red contours) integrated over the LSR velocity ranges of 5.5–7 (blue) and 11.5–13 (red) km s^{-1} . The contours start at 3σ and increase by 3σ , where σ is the rms noise in each integrated map. The large dashed circle shows the primary beam of the SMA at 230 GHz, centred on the phase centre. The synthesized beam is shown in the lower left corner of each panel, and the crosses mark the positions of the protostars. The solid lines in each panel indicate the approximate morphologies of the outflows we have identified as being driven by each of the three protostars. These lines are not fits, they are drawn by hand and are only meant to serve as a rough guide.

Per-emb 53. Per-emb 53 is clearly associated with a bipolar molecular outflow. Our method accurately recovers the opening angle of this outflow close to the driving source, where the outflow is widest. We thus place this object in the HC group.

Per-emb 56. The outflow driven by Per-emb 56 is clearly detected and well fitted by our method, thus we place this object in the HC group.

Per-emb 57. A collimated, jet-like outflow feature is detected south of the Per-emb 57 protostar, extending on a line that traces back to the protostar. Assuming this emission is indeed associated with an outflow driven by Per-emb 57, our method accurately fits the morphology of this jet-like feature. However, the fact that this feature does not extend all the way back to the protostar, coupled with the presence of only weak emission to the northwest that is not consistent with being the counterpart to this feature, lead us to place this object in the low confidence group.

Per-emb 61. A bipolar north–south outflow driven by Per-emb 61 is detected in our *MASSES* data. While each lobe appears fairly collimated at first glance, the position angles of the two lobes appear misaligned. An alternative interpretation is that the outflow is wider than implied by either lobe individually, with only one of the two cavity walls detected in each lobe. We favour this second interpretation based on the detection, in individual velocity channels, of weak, patchy emission where the other cavity walls would be present. Since we cannot fully rule out a more collimated outflow with misaligned lobes, we place this object in the MC group.

Per-emb 62. The outflow driven by Per-emb 62 is clearly detected, although only one of the two cavity walls appears to be detected in the southern lobe. As this outflow is well fitted by our method, we place this object in the HC group.

B1-bS. B1-bS, either a very young Class 0 protostar or a first hydrostatic core, is located in the B1-b system (Pezzuto et al. 2012) and drives a southeast (blueshifted)–northwest (redshifted) bipolar outflow, with the redshifted outflow not detected in CO due to confusion with emission from other components in the B1-b system but detected in other tracers (Hirano & Liu 2014; Gerin et al. 2015; Lee et al. 2016). Since only the blueshifted lobe can be clearly identified and separated in our ^{12}CO (2 – 1) observations, we only apply our fitting method to this lobe. While the robust +1 image only marginally resolves the length of this outflow and does not resolve

its width, the sensitivity in the robust –1 image is too low to obtain a reliable fit. Thus we apply our fitting method to the robust +1 image and present our opening angle as an upper limit.

L1448N-NW. L1448N-NW is part of the Per-emb 33 system (also referred to as the L1448N system or the L1448 IRS 3 system), a hierarchical system of six protostars distributed in a close triple system (L1448N-B), a close binary system (L1448N-A), and a single system (L1448N-NW, Lee et al. 2015; Tobin et al. 2016). The *MASSES* ^{12}CO (2 – 1) observations show a complicated tangle of outflow emission from the six protostars in this system, as well as from the nearby Per-emb 26 outflow (see Fig. A5; Lee et al. 2015). Despite the confusion in this region the outflow from L1448N-NW can be clearly identified and separated from the other emission components, with a southeast redshifted lobe and a northwest blueshifted lobe. As the driving source is near the edge of the primary beam the blueshifted lobe (which extends away from the phase centre) is not well detected, thus we only fit to the redshifted lobe. We still include this outflow in the HC group due to its simply morphology and the excellent visual match between the fit and the image. We note that L1448N-NW is actually a binary system with a projected separation of 0.25 arcsec (corresponding to a projected linear separation of 75 au; Tobin et al. 2016, 2018). As we see no evidence of multiple outflows, we assume that one component dominates both the outflow and the observed SED.

L1448N-B. L1448N-B is part of the Per-emb 33 system (also referred to as the L1448N system or the L1448 IRS 3 system), a hierarchical system of six protostars distributed in a close triple system (L1448N-B), a close binary system (L1448N-A), and a single system (L1448N-NW, Lee et al. 2015; Tobin et al. 2016). The *MASSES* ^{12}CO (2 – 1) observations show a complicated tangle of outflow emission from the six protostars in this system, as well as from the nearby Per-emb 26 outflow (see Fig. A5; Lee et al. 2015). Despite this complexity, we are able to clearly identify an outflow driven by the L1448N-B triple system. The southeastern redshifted lobe is particularly clear, whereas the northwestern blueshifted lobe is likely contaminated by blueshifted emission from the Per-emb 26 outflow (see Fig. A5). Thus, we fit only the redshifted lobe, and assume that both the outflow and the observed SED are dominated by the same single member of this triple system. Given the uncertainties in these assumptions, we place this outflow in the MC group.

Per-Bolo 45. Per-Bolo 45 was identified as a candidate first hydrostatic core based first on the detection of compact 3 mm continuum emission (Schnee et al. 2010) and then on the tentative detection of an outflow in SiO (2 – 1) emission (Schnee et al. 2012). We detect blueshifted ^{12}CO (2 – 1) emission to the southeast of the compact continuum source that is broadly consistent with the location, morphology, and velocities of the outflow detection claimed by Schnee et al. (2012). While this blueshifted feature does not extend all the way back to the position of the continuum source, the long axis of the extended morphology does trace back to the continuum source. We do not detect a corresponding redshifted feature to the northwest, but this could possibly be due to the significant offset between the phase centre of the observations and the position of the compact continuum source. While the general agreement with the claimed outflow by Schnee et al. (2012) is encouraging, the highly tentative nature of this detection leads us to place this object in the low confidence group, a point reinforced by the fact that Maureira et al. (2020) argued in favour of a prestellar nature for this object.

Per-Bolo 58. Our MASSES data detect a known bipolar outflow driven by Per-Bolo 58, first identified as a candidate first hydrostatic core (Enoch et al. 2010), but currently believed to be a very young Class 0 protostar (Maureira et al. 2020). Our outflow detection matches that of Dunham et al. (2011). Additional blueshifted emission detected to the north of the redshifted lobe, and to the south of the blueshifted lobe, likely arises from the outflow driven by Per-emb 23 and is excluded from our analysis. Given the extent to which our fit agrees visually with the morphology of this outflow, we place this object in the HC group.

SVS 13C. SVS 13C is part of the SVS 13 system, located approximately 20 arcsec (6000 au) southwest of SVS 13A and approximately 9 arcsec (2700 au) southwest of SVS 13B (Looney et al. 2000). Plunkett et al. (2013) identified a tentative outflow extending approximately north–south, with the southern lobe associated with weak blueshifted emission that aligns with a scattered light cavity seen in a 4.5 μm image from the *Spitzer Space Telescope*. The counter lobe extending to the north is less clearly detected, although there is substantial confusion in the CO emission in this region (see Plunkett et al. 2013, for details). In our MASSES data we detect blueshifted emission extending north and south of SVS 13C. To the south, this emission aligns well with the scattered light cavity seen in the *Spitzer* data (see, in particular, fig. 8 of Plunkett et al. 2013), and with the axis of the much larger scale outflow identified by Plunkett et al. We also detect redshifted emission to the northwest of the protostar, but like Plunkett et al. are unable to conclusively identify the origin of this emission. We fit only the blueshifted emission, but given the substantial uncertainties in interpreting this complicated environment, we place this object in the low confidence group.

APPENDIX C: NOTES ON INDIVIDUAL SOURCES WITHOUT MEASURED OPENING ANGLES

In this appendix, we provide relevant notes on each source in the MASSES sample for which we are unable to obtain opening angle measurements. In general, we do not show images for these objects and instead refer to Stephens et al. (2018, 2019), where the full set of MASSES ^{12}CO (2 – 1) images are displayed.

Per-emb 4. Per-emb 4 was identified as a protostar by Enoch et al. (2009) based on the association of a mid-infrared *Spitzer* source with a 1.1 mm Bolocam continuum detection. However, as first noted

by Stephens et al. (2018), the protostellar status of this object is questionable due to its non-detection in our MASSES 230 GHz continuum observations (Pokhrel et al. 2018), our MASSES C ^{18}O (2 – 1) observations (Stephens et al. 2018), and in the VANDAM 8 mm continuum observations (Tobin et al. 2016). While we detect some redshifted emission in the vicinity of Per-emb 4, up to velocities of at least 5 km s $^{-1}$ from the systemic cloud velocity, we do not detect any clear signatures of an outflow associated with this object. We thus tentatively conclude that there is no outflow driven by this object detected in our data.

Per-emb 8/55. Per-emb 8 and Per-emb 55 form a wide binary system, with a projected separation of 9.6 arcsec (corresponding to a projected linear separation of 2867 au; Tobin et al. 2016, 2018). Per-emb 55 is additionally a close binary system, with a projected linear separation of 185 au. Fig. B4 plots contours displaying the integrated blueshifted and redshifted ^{12}CO (2 – 1) emission over LSR (local standard of rest) velocity ranges of 5.5–7 and 11.5–13 km s $^{-1}$, respectively. Careful inspection of these contours, along with individual channel maps (not shown), reveals substantial but ultimately inconclusive evidence for three distinct outflows in this system (see also the higher resolution ^{12}CO (2 – 1) ALMA maps presented by Tobin et al. 2018). Lines are drawn in Fig. B4 to indicate the approximate opening angles of these outflows. These lines are drawn by eye and should not be interpreted as fits to the data. While these outflows do generally agree with the detections reported previously by Walawender et al. (2006), we are unable to obtain even low confidence fits given the extensive degree to which they overlap with each other. Thus we do not include these outflows in our analysis.

Per-emb 12. Per-emb 12 (NGC1333 IRAS 4A) is a binary system with a projected angular separation of 1.8 arcsec, corresponding to a projected linear separation of 549 au (e.g. Tobin et al. 2016, 2018). Both components drive outflows that overlap and likely interact with each other (e.g. Santangelo et al. 2015; Ching et al. 2016). As these two components are impossible to fully disentangle due to their overlapping nature, and furthermore given that the component that dominates the large-scale outflow emission is not the component that dominates the observed infrared and (sub)millimetre SED (Looney et al. 2000; Reipurth et al. 2002; Enoch et al. 2009; Santangelo et al. 2015), we are unable to reliably measure opening angles and bolometric temperatures for either component.

Per-emb 14. No outflow was detected from Per-emb 14. Similarly, no outflow was detected from this target in the large-scale map of NGC 1333 outflows presented by Plunkett et al. (2013).

Per-emb 18. Per-emb 18 is part of the Per-emb 18/21 wide binary system, with a projected angular separation of 13.3 arcsec (corresponding to a projected linear separation of 3976 au; Tobin et al. 2016, 2018). We do detect a clear bipolar outflow driven by Per-emb 21 (see also Lee et al. 2016), but not for Per-emb 18. While there is some evidence for such an outflow in the MASSES ^{12}CO (3–2) data and *Spitzer-IRAC* 4.5 μm image (Lee et al. 2016), as well as in higher resolution ^{12}CO (2 – 1) ALMA maps presented by Tobin et al. (2018), we are unable to clearly identify, separate, and resolve this outflow.

Per-emb 32. Per-emb 32, located within IC348, is a binary system with a projected separation of 6 arcsec (corresponding to a projected linear separation of 1820 au, Evans et al. 2009; Dunham et al. 2015; Young et al. 2015; Tobin et al. 2016, 2018). Per-emb 32 is associated with both blueshifted and redshifted ^{12}CO (2 – 1) emission at velocities exceeding 3 km s $^{-1}$ relative to the systemic cloud velocity, and while this emission is not clearly associated with other outflows in IC348 (see Fig. A1), we are unable to clearly

identify outflows from one or both components of this binary system.

Per-emb 35. Per-emb 35, located near the western edge of NGC 1333, is a binary system with a projected separation of 1.9 arcsec (corresponding to a projected separation of 572 au; Tobin et al. 2016, 2018). Tobin et al. (2018) present evidence for approximately parallel southeast–northwest outflows driven by the two components of this binary system using higher resolution ^{12}CO (2–1) ALMA data, but we are unable to adequately resolve these separate outflows in our MASSES data. Thus, despite the likely presence of one or more outflows, we are unable to conclusively measure position and opening angles for this system.

Per-emb 37. Per-emb 37, located in the northeastern portion of NGC 1333, is a wide binary system with a projected separation of 10.6 arcsec (corresponding to a projected linear separation of 3167 au; Young et al. 2015; Tobin et al. 2016, 2018). Both components are detected in the VANDAM 8 mm continuum observations (Tobin et al. 2016), but only one is detected in the MASSES 1.3 mm continuum observations (Pokhrel et al. 2018; Stephens et al. 2018). While there is clearly high-velocity emission detected in the MASSES ^{12}CO (2–1) observations, the complexity of this region prevents us from clearly identifying individual outflows.

Per-emb 38. Per-emb 38, located 11 arcmin (0.96 pc) southwest of the B1 region, is single down to a projected separation of at least 20 au (Tobin et al. 2016, 2018). This object is associated with high-velocity ^{12}CO (2–1) emission in our MASSES data, and given the isolated nature of this protostar it is highly likely that the emission originates in an outflow driven by Per-emb 38. However, we are unable to clearly identify distinct outflow lobes.

Per-emb 39. Per-emb 39 is not detected in the 1.3 mm continuum or C^{18}O (2–1) MASSES observations (Stephens et al. 2018), nor is it detected in the VANDAM 8 mm continuum (Tobin et al. 2016). It also has an extremely low and highly uncertain bolometric luminosity (Enoch et al. 2009; Stephens et al. 2018). For these reasons, Stephens et al. (2018) questioned the protostellar status of this object. While we detect ^{12}CO (2–1) emission in the vicinity of Per-emb 39 in our MASSES data up to $\sim 5 \text{ km s}^{-1}$ from rest, particularly at redshifted velocities, we are unable to conclusively identify the origin of this emission. Furthermore, while Hatchell & Dunham (2009) claim the detection of an outflow associated with this object, the claim is based only on the presence of high-velocity emission, and indeed their Fig. 3 shows no clear morphological evidence for an outflow. We are thus unable to clearly identify and fit to an outflow.

Per-emb 43. Per-emb 43 was identified as a protostar by Enoch et al. (2009) based on the association of a mid-infrared *Spitzer* source with a 1.1 mm Bolocam continuum detection. However, as first noted by Stephens et al. (2018), the protostellar status of this object is questionable due to its non-detection in our MASSES 230 GHz continuum observations (Pokhrel et al. 2018), our MASSES 230 GHz C^{18}O (2–1) observations (Stephens et al. 2018), and in the 8 mm VANDAM observations (Tobin et al. 2016). We see no evidence for an outflow in our MASSES ^{12}CO (2–1) observations (see also fig. 3 of Stephens et al. 2018, for a similar result).

Per-emb 45. Per-emb 45 was identified as a protostar by Enoch et al. (2009) based on the association of a mid-infrared *Spitzer* source with a 1.1 mm Bolocam continuum detection. However, similar to Per-emb 43, the status of Per-emb 45 as a protostar was noted as questionable by Stephens et al. (2018) due to its non-detection in our MASSES 230 GHz continuum observations (Pokhrel et al. 2018), our MASSES 230 GHz C^{18}O (2–1) observations (Stephens et al. 2018), and in the 8 mm VANDAM observations (Tobin et al. 2016).

We see no evidence for an outflow in our MASSES ^{12}CO (2–1) observations, as seen in Fig. A2.

Per-emb 47. We detect substantial redshifted emission in the vicinity of Per-emb 47 in our MASSES ^{12}CO (2–1) observations, extending up to $\sim 10 \text{ km s}^{-1}$ from rest. While this emission is likely associated with an outflow, as seen by comparing to the single-dish maps presented by Hatchell & Dunham (2009, see in particular their fig. 5), the morphology of the emission detected by MASSES is not clearly indicative of an outflow (see Section 5.3.2 for further discussion of this object). We thus do not include this object in our analysis.

Per-emb 48. Per-emb 48, located in the L1455 region of Perseus (see Fig. A4), is a binary system with a projected angular separation of 0.35 arcsec (corresponding to a projected linear separation of 104 au; Tobin et al. 2016, 2018). While we detect ^{12}CO (2–1) emission at velocities exceeding 3 km s^{-1} from the systemic velocity, we are unable to identify clear signatures of one or more outflows driven by this system. There is also no outflow clearly associated with this system in the single-dish maps presented by Curtis et al. (2010). Furthermore, we note that the blueshifted lobe of the Per-emb 20 outflow is pointed toward Per-emb 48 (see Fig. A4), rendering it unclear if the blueshifted emission in the vicinity of Per-emb 48 is actually associated with this system or with Per-emb 20.

Per-emb 49. Per-emb 49 is a binary system with a projected angular separation of 0.313 arcsec (corresponding to a projected linear separation of 94 au; Tobin et al. 2016, 2018). While undetected in our MASSES 1.3 mm continuum observations, we do detect evidence for a northeast–southwest outflow driven by this system. However, as our map is dominated by sidelobe emission from the nearby Per-emb 21 outflow (see Fig. A3), we are unable to reliably fit for the opening and position angles of this outflow.

Per-emb 51. We do not detect any evidence of an outflow driven by Per-emb 51 in our MASSES ^{12}CO (2–1) observations.

Per-emb 54. Per-emb 54, located in NGC 1333, is detected as a single source by the VANDAM survey at 8 mm (Tobin et al. 2016) but is not associated detected in our MASSES 230 GHz continuum observations (Pokhrel et al. 2018; Stephens et al. 2018). We detect high-velocity ^{12}CO (2–1) emission in our MASSES observations, extending up to $\sim 10 \text{ km s}^{-1}$ from the systemic velocity. This emission, which is also detected at much lower resolution by Hatchell et al. (2007) and Curtis et al. (2010), does not clearly originate from any other object in NGC 1333 and is thus likely indicative of an outflow driven by Per-emb 54. However, we are unable to identify any clear morphological signatures of an outflow.

Per-emb 58/65. Per-emb 58 and Per-emb 65, located in NGC 1333, are separated by 29 arcsec (corresponding to a projected linear separation of 8700 au; Tobin et al. 2016, 2018). Additionally, there are several Class II YSOs (Young Stellar Objects) in the vicinity of these two protostars (e.g. Young et al. 2015). While there is copious high-velocity ^{12}CO (2–1) emission detected in the vicinity of these objects (see e.g. fig. 3 from Stephens et al. 2018), indicating the likely presence of one or more outflows, we are unable to definitively identify individual outflows or assign driving sources.

Per-emb 59. Per-emb 59 was identified as a protostar by Enoch et al. (2009) based on the association of a mid-infrared *Spitzer* source with a 1.1 mm Bolocam continuum detection. However, as first noted by Stephens et al. (2018), the protostellar status of this object is questionable due to its non-detection in our MASSES 230 GHz

continuum observations (Pokhrel et al. 2018), our MASSSES 230 GHz C^{18}O ($2-1$) observations (Stephens et al. 2018), and in the 8 mm VANDAM observations (Tobin et al. 2016). We see no evidence for an outflow in our MASSSES ^{12}CO ($2-1$) observations (see also fig. 3 of Stephens et al. 2018, for a similar result).

Per-emb 60. Per-emb 60 is not detected in the 1.3 mm continuum or C^{18}O ($2-1$) MASSSES observations (Stephens et al. 2018), nor is it detected in the VANDAM 8 mm continuum (Tobin et al. 2016). It also has an extremely low and highly uncertain bolometric luminosity (Enoch et al. 2009; Stephens et al. 2018). For these reasons, Stephens et al. (2018) questioned the protostellar status of this object, although they did note the presence of relatively compact emission in single-dish continuum maps (e.g. Suresh et al. 2016). We do not detect any evidence for an outflow driven by this object.

Per-emb 63. While we detect high-velocity ^{12}CO ($2-1$) emission in the vicinity of Per-emb 63, we do not detect any conclusive morphological evidence for an outflow driven by this object.

Per-emb 64. While we detect high-velocity ^{12}CO ($2-1$) emission in the vicinity of Per-emb 64, we do not detect any conclusive morphological evidence for an outflow driven by this object.

Per-emb 66. Per-emb 66, identified as a protostar by Enoch et al. (2009), is not associated with continuum detections by MASSSES at 1.3 mm (Pokhrel et al. 2018; Stephens et al. 2018) or by VANDAM at 8 mm (Tobin et al. 2016). We detect a compact structure in ^{12}CO ($2-1$) that could be tracing either a compact outflow or a large, rotating structure (see also fig. 3 of Stephens et al. 2018), but we are unable to distinguish between these possibilities. Even if we assumed it is tracing an outflow, all of the emission is within the 4 arcsec radius from the driving source over which we mask emission to avoid excessive beam-smearing effects.

B1-bN. B1-bN, first identified by Pezzuto et al. (2012), is either a very young Class 0 protostar or a first hydrostatic core. Gerin et al. (2015) detected a compact, east–west outflow driven by B1-bN in their methanol and formaldehyde observation, and while we see signs of this outflow in our ^{12}CO ($2-1$) observations, we are unable to clearly identify and separate this outflow in either position or velocity.

L1448 IRS2E. L1448 IRS2E was first identified as a candidate first hydrostatic core by Chen et al. (2010), who detected a compact 1.3 mm continuum source and a redshifted, jet-like feature in ^{12}CO ($2-1$) emission. This redshifted feature is located within the redshifted lobe of the nearby Per-emb 22 outflow (see Fig. A5) and is detected up to velocities of 25 km s^{-1} (much too fast for outflows driven by first hydrostatic cores; e.g. Machida, Inutsuka & Matsumoto 2008; Price, Tricco & Bate 2012; Dunham et al. 2014a), leading to substantial uncertainty in the nature of this object. While we detect this redshifted emission in our MASSSES ^{12}CO ($2-1$) observations, we do not detect the compact source in our 1.3 mm continuum observations even though our sensitivity should be sufficient given the flux density reported by Chen et al. (2010). We thus agree with Maureira et al. (2020) that there is no compact source driving an outflow at this position.

L1451-mm. L1451-mm was first identified by Pineda et al. (2011) as a candidate first hydrostatic core, and current evidence confirms it is one of the best known candidates for a hydrostatic core (Maureira et al. 2017, 2020). We detect a very slow, very compact outflow consistent with Pineda et al. (2011, see their fig. 7). However, as this outflow is essentially unresolved and contained entirely within 4 arcsec of its driving source, we are unable to reliably measure opening and position angles.

SVS 13B. SVS 13B drives a collimated, jet-like outflow detected in SiO emission (Bachiller et al. 1998), with the blueshifted lobe

extending to the south and the redshifted lobe extending to the north. While we do see evidence for this same outflow in our MASSSES ^{12}CO ($2-1$) data, the fact that it is located on the edge of the bright, blueshifted lobe of the outflow driven by Per-emb 44 (SVS 13A) makes it impossible to accurately separate and fit this outflow.

APPENDIX D: CORRECTIONS FOR SOURCE INCLINATION AND MEASUREMENT TECHNIQUE

D1 Corrections for source inclination

Here, we describe our method for correcting an outflow opening angles, θ_c , for source inclination (or, alternatively, for projecting a corrected opening angle back onto the plane of the sky). This method assumes conical shaped outflow cavities.

As seen in the left panel of Fig. D1, for a conical outflow cavity,

$$\tan\left(\frac{\theta_c}{2}\right) = \frac{(\text{outflow half-width})}{(\text{outflow extent})}. \quad (\text{D1})$$

Additionally, as seen in the right panel of Fig. D1, for an observer viewing the outflow at an inclination i ,

$$(\text{observed extent}) = (\text{real extent}) \sin(i), \quad (\text{D2})$$

Since the outflow extent is reduced by a factor of $\sin(i)$ but the outflow half-width is unchanged, the observed opening angle, θ_c^{obs} , is given by

$$\tan\left(\frac{\theta_c^{\text{obs}}}{2}\right) = \frac{(\text{real half-width})}{(\text{observed extent})} = \frac{(\text{real half-width})}{(\text{real extent})} \frac{1}{\sin(i)}. \quad (\text{D3})$$

Comparing equations (D1) and (D3), the observed and intrinsic outflow opening angles are related by

$$\tan\left(\frac{\theta_c}{2}\right) = \tan\left(\frac{\theta_c^{\text{obs}}}{2}\right) \sin(i). \quad (\text{D4})$$

Equation (D4) can be used to either correct the observed θ_c^{obs} for inclination, or to project an inclination-corrected opening angle θ_c back onto the plane of the sky.

D2 Corrections for measurement technique

In two of the previous studies to which we compare, namely Hsieh et al. (2017) and Habel et al. (2021), curved outflow cavities given by power-law functions are fit to the cavity morphologies detected in infrared light. Since the opening angle of such an outflow cavity depends on the height above the mid-plane where it is measured (wider near the mid-plane and narrower farther from the mid-plane), both studies tabulate the opening angles of cones that intersect the power-law cavity walls at fixed vertical heights above the mid-plane. The specific vertical heights chosen in each study are 5000 au for Hsieh et al. (2017) and 8000 au for Habel et al. (2021).

The left panel of Fig. D2 shows an example of this process for a power-law cavity given by:

$$y = Ax^p, \quad (\text{D5})$$

where $A = 0.02$ and $p = 1.5$ in this specific example. This example is meant to illustrate the process as it was followed by Hsieh et al.

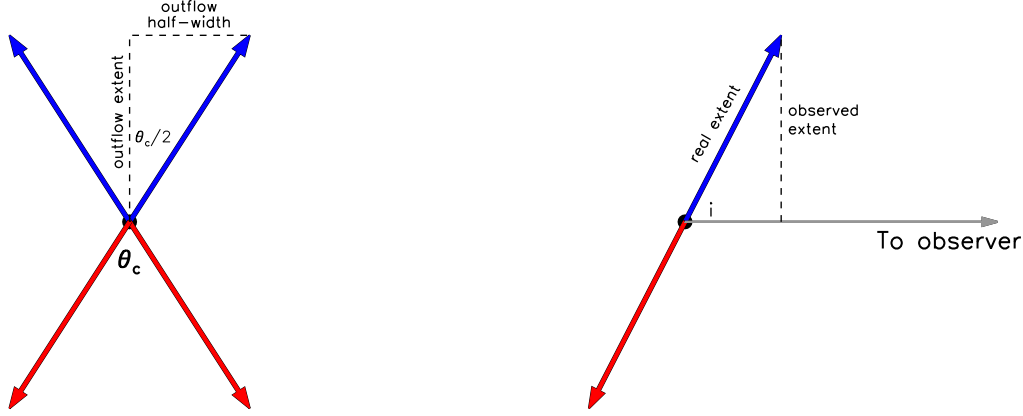


Figure D1. Left: geometry of an outflow viewed edge-on. Right: geometry of the real and observed extents of an outflow when viewed at an inclination angle i .

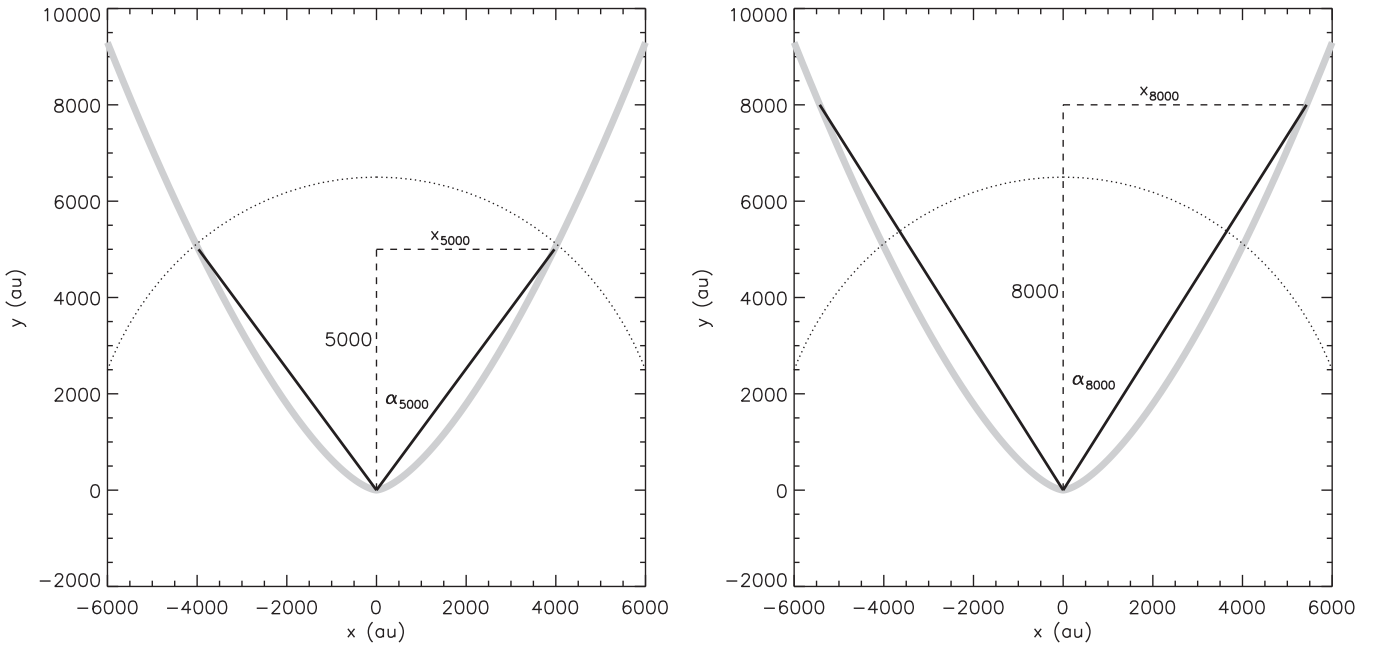


Figure D2. Comparison of curved (power law) and conical outflow cavity shapes. In both panels, the dotted line shows an assumed core boundary of 6500 au (see the text for details) and the thick grey line shows a power-law cavity given by $y = Ax^p$, where $A = 0.02$ and $p = 1.5$ in this specific example. The solid black lines show cones that intersect the power-law cavity walls at fixed vertical heights above the mid-plane (5000 au in the left panel and 8000 au in the right panel).

(2017), who also adopted $p = 1.5$ in their fits. For the specific example shown in this figure, the opening half-angle is given by:

$$\alpha_{5000} = \arctan\left(\frac{x_{5000}}{5000}\right) = \arctan\left(\frac{3968.5}{5000}\right) = 38.4^\circ, \quad (\text{D6})$$

giving a full opening angle of 76.8° (we note here that we use θ to refer to full opening angles and α to refer to half opening angles).

This method for measuring outflow opening angles is very different than the method used by us (and by Hsieh et al. 2023). In an appendix to their paper, Myers et al. (2023) compared the two methods and found that they show excellent statistical agreement, with the differences between the two methods smaller than the typical 5° – 15° uncertainties in the method used in this paper. Myers et al. (2023) performed this comparison using cones that intersect the power-law cavity walls at a fixed radius of 6500 au rather than a fixed vertical height of 5000 au. As seen in Fig. D2, differences between

these two cones are minimal (a cone drawn to the intersection of the 6500 au core boundary and the power-law cavity shape would be almost identical to the cone shown in the figure). Thus, we do not perform any corrections for measurement technique before comparing our measurements with opening angles measured by Hsieh et al. (2017).

However, as seen in the right panel of Fig. D2, the vertical height of 8000 au used by Habel et al. (2021) results in narrower cones. To ensure an apples-to-apples comparison between their results and ours, we thus correct their reported opening half-angles (α_{8000}) to those measured at a vertical height of 5000 au (α_{5000}). To perform this correction, we first solve equation (D5) for A to calculate the specific value of A for each outflow in the Habel et al. (2021) sample as follows:

$$A = \frac{y}{x^p} = \frac{y}{[y \tan(\alpha)]^p} = \frac{8000}{[8000 \tan(\alpha_{8000})]^p}. \quad (\text{D7})$$

Once the value of A is known, we then solve equation (D5) for x to calculate the horizontal coordinate where the power-law cavity reaches a vertical height of 5000 au:

$$x_{5000} = \left(\frac{y}{A}\right)^{\frac{1}{p}} = \left(\frac{5000}{A}\right)^{\frac{1}{p}}. \quad (\text{D8})$$

Finally, we calculate α_{5000} as:

$$\alpha_{5000} = \arctan\left(\frac{x_{5000}}{y}\right) = \arctan\left(\frac{x_{5000}}{5000}\right). \quad (\text{D9})$$

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